

# CERTIFICATE

This is to certify that the project titled “**SPATIO-FREQUENCY ANALYSIS OF FACIAL REGIONS FOR VIDEO FORGERY DETECTION**” is a genuine work submitted by **Shiv Vendra Singh (23102012)** and **Annanya Priyadarshini Sahoo (23102200)**, students of the Bachelor of Technology degree in Electronics and Communication Engineering at Jaypee Institute of Information Technology, Noida, under the guidance and supervision of **Prof. Vineet Khandelwal**.

The project has been carried out as a part of the Minor Project (6th Semester) in partial fulfilment of the requirements for the award of the Bachelor of Technology degree. This work is an original piece of research and has not been submitted elsewhere for any other purpose. It is hereby confirmed that the work reported in this project is original and was performed under my supervision. We also acknowledge that this project has been reviewed and approved for presentation.

**Vineet Khandelwal**

Professor

Department of Electronics and Commu-  
nication Engineering

JIIT Noida

**Jitendra Mohan**

Associate Head

Department of Electronics and  
Communication Engineering

JIIT Noida

# DECLARATION

We hereby declare that the Minor Project Report entitled “**SPATIO-FREQUENCY ANALYSIS OF FACIAL REGIONS FOR VIDEO FORGERY DETECTION**”, submitted to Jaypee Institute of Information Technology in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering, is a record of our original work carried out under the supervision of **Prof. Vineet Khandelwal**.

We further declare that this report has not been submitted to any other university or institute for the award of any degree or diploma.

**Date: April 28, 2026**

**Names:**

Shiv Vendra Singh (23102012)

Annanya Priyadarshini Sahoo (23102200)

**Under the Supervision of:**

Prof. Vineet Khandelwal

# ACKNOWLEDGEMENT

If words are considered a symbol of approval and a token of appreciation, then let these words play the heralding role in expressing our gratitude. The satisfaction that accompanies the successful completion of any task would be incomplete without acknowledging the people whose ceaseless cooperation made it possible, and whose constant guidance and encouragement crown all efforts with success.

We are deeply grateful to our project mentor, **Prof. Vineet Khandelwal**, for his guidance, inspiration, and constructive suggestions throughout the course of this project. His insights, new ideas, and consistent encouragement helped us navigate multiple technical and conceptual aspects of our work, enabling us to improve the quality and depth of this project.

We also extend our heartfelt thanks to our institution, Jaypee Institute of Information Technology, and to all faculty members of the Department of Electronics and Communication Engineering, whose support and academic environment made this project a reality.

# Contents

|                                                                          |           |
|--------------------------------------------------------------------------|-----------|
| <b>List of Figures</b>                                                   | <b>6</b>  |
| <b>List of Tables</b>                                                    | <b>7</b>  |
| <b>1 Overview of the Project</b>                                         | <b>8</b>  |
| Chapter Introduction . . . . .                                           | 8         |
| 1.1 Motivation . . . . .                                                 | 9         |
| 1.2 Literature Review . . . . .                                          | 10        |
| 1.2.1 Review of Existing Deepfake Detection Approaches . . . . .         | 10        |
| 1.2.2 Synthesis of Literature and Identified Research Gaps . . . . .     | 14        |
| 1.3 Existing Challenges . . . . .                                        | 15        |
| 1.4 Objectives of the Project . . . . .                                  | 15        |
| 1.5 Organisation of the Report . . . . .                                 | 17        |
| <b>2 Foundational Study</b>                                              | <b>18</b> |
| Chapter Introduction . . . . .                                           | 18        |
| 2.1 Background Study . . . . .                                           | 18        |
| 2.1.1 Fundamentals of Video Forgery Detection . . . . .                  | 18        |
| 2.1.2 Feature Extraction and Preprocessing Techniques . . . . .          | 20        |
| 2.1.3 Deep Learning Models and Integrated Detection Approaches . . . . . | 22        |
| Chapter Summary . . . . .                                                | 23        |
| <b>3 System Design and Methodology</b>                                   | <b>24</b> |
| 3.1 Introduction . . . . .                                               | 24        |
| 3.2 System Design . . . . .                                              | 25        |
| 3.2.1 Design Philosophy and Motivation . . . . .                         | 25        |
| 3.2.2 Dataset: FaceForensics++ . . . . .                                 | 26        |
| 3.2.3 Data Partitioning Strategy . . . . .                               | 26        |
| 3.2.4 Frame Extraction . . . . .                                         | 27        |
| 3.2.5 Face Detection and Alignment . . . . .                             | 28        |
| 3.2.6 Class Imbalance Mitigation . . . . .                               | 28        |
| 3.3 Model Illustration . . . . .                                         | 29        |
| 3.3.1 Overview of the SWTVIT Architecture . . . . .                      | 29        |
| 3.3.2 Stationary Wavelet Transform (SWT) Preprocessing . . . . .         | 29        |

|          |                                                                |           |
|----------|----------------------------------------------------------------|-----------|
| 3.3.3    | Vision Transformer Backbone . . . . .                          | 30        |
| 3.3.4    | Classification Head . . . . .                                  | 32        |
| 3.3.5    | GradCAM-based Spatial Localization . . . . .                   | 32        |
| 3.3.6    | Data Augmentation . . . . .                                    | 33        |
| 3.4      | Working Principle . . . . .                                    | 34        |
| 3.4.1    | Training Configuration . . . . .                               | 34        |
| 3.4.2    | Inference Pipeline . . . . .                                   | 35        |
| 3.4.3    | Real-World Generalization Testing . . . . .                    | 36        |
| 3.4.4    | Checkpoint Management and Reproducibility . . . . .            | 37        |
| 3.5      | Chapter Summary . . . . .                                      | 37        |
| <b>4</b> | <b>Results and Discussion</b>                                  | <b>39</b> |
| 4.1      | Introduction . . . . .                                         | 39        |
| 4.2      | Dataset Description . . . . .                                  | 39        |
| 4.2.1    | Dataset Overview . . . . .                                     | 39        |
| 4.2.2    | Frame and Face Extraction Statistics . . . . .                 | 40        |
| 4.2.3    | Dataset Partition Summary . . . . .                            | 41        |
| 4.3      | Parameters Used . . . . .                                      | 42        |
| 4.3.1    | Model Architecture Parameters . . . . .                        | 42        |
| 4.3.2    | Training Hyperparameters . . . . .                             | 43        |
| 4.3.3    | Preprocessing and Augmentation Parameters . . . . .            | 44        |
| 4.4      | Comparison With Existing Techniques . . . . .                  | 46        |
| 4.4.1    | Positioning Within the Literature . . . . .                    | 46        |
| 4.4.2    | Discussion of Comparative Results . . . . .                    | 47        |
| 4.5      | Results . . . . .                                              | 48        |
| 4.5.1    | Training Dynamics . . . . .                                    | 48        |
| 4.5.2    | Frame-Level Test Results . . . . .                             | 49        |
| 4.5.3    | Frame-Level Confusion Matrix . . . . .                         | 50        |
| 4.5.4    | Video-Level Test Results . . . . .                             | 51        |
| 4.5.5    | Video-Level Confusion Matrix . . . . .                         | 52        |
| 4.5.6    | GradCAM Spatial Localization Results . . . . .                 | 52        |
| 4.5.7    | Corresponding Video Set Evaluation . . . . .                   | 54        |
| 4.5.8    | Real-World Generalization: Personal Video Evaluation . . . . . | 55        |
| 4.5.9    | Final Pipeline Summary . . . . .                               | 56        |
| 4.5.10   | Consolidated Results Summary . . . . .                         | 57        |
| <b>5</b> | <b>Conclusion and Future Scope</b>                             | <b>59</b> |
|          | Chapter Introduction . . . . .                                 | 59        |
| 5.1      | Conclusion . . . . .                                           | 59        |
| 5.2      | Future Scope . . . . .                                         | 60        |
|          | References . . . . .                                           | 63        |

# List of Figures

|     |                                                                                                                                                                               |    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 4.1 | Aligned MTCNN face crops ( $224 \times 224$ px) from each class: (a) Real, (b) Deepfake, (c) FaceSwap, (d) Face2Face. . . . .                                                 | 41 |
| 4.2 | Training and validation loss (left) and accuracy (right) curves over 24 epochs. . . . .                                                                                       | 48 |
| 4.3 | Frame-level confusion matrices on the FF++ test set (1,501 frames): raw counts (left) and row-normalized (right). . . . .                                                     | 50 |
| 4.4 | Video-level confusion matrices on the FF++ test set (80 videos): raw counts (left) and row-normalized (right). . . . .                                                        | 52 |
| 4.5 | Terminal output from the personal video generalization evaluation. Top: genuine video classified as REAL. Bottom: face-swap video classified unanimously as FACESWAP. . . . . | 55 |
| 4.6 | Final SWTVIT pipeline summary: frame-level accuracy 84.61%, video-level accuracy 90.00%, macro F1 85.09%, best validation accuracy 84.81% at epoch 16. . . . .                | 56 |

# List of Tables

|      |                                                                                                                           |    |
|------|---------------------------------------------------------------------------------------------------------------------------|----|
| 4.1  | FaceForensics++ dataset composition (c23 format, 125 videos per class). . .                                               | 40 |
| 4.2  | Frame and face counts per class (100% MTCNN detection rate across all splits). . . . .                                    | 41 |
| 4.3  | Video-level stratified dataset partition summary. . . . .                                                                 | 42 |
| 4.4  | ViT-Small architectural parameters used in SWTVIT. . . . .                                                                | 43 |
| 4.5  | SWTVIT training hyperparameters. . . . .                                                                                  | 44 |
| 4.6  | Preprocessing and data augmentation parameters for the SWTVIT pipeline.                                                   | 45 |
| 4.7  | Comparison of deepfake detection methods on FF++ (c23). † = frame-level accuracy; * = binary classification only. . . . . | 46 |
| 4.8  | Frame-level classification metrics on the FF++ test set (1,501 frames, epoch-16 checkpoint). . . . .                      | 49 |
| 4.9  | Video-level classification metrics on the FF++ test set (80 videos, 20 per class). . . . .                                | 51 |
| 4.10 | Corresponding video evaluation results for subject 585 (all four videos correctly classified). . . . .                    | 54 |
| 4.11 | Real-world generalization results on independently produced personal videos.                                              | 55 |
| 4.12 | Consolidated SWTVIT performance metrics at frame and video level on the FF++ test set. . . . .                            | 57 |

# Chapter 1 Overview of the Project

## Introduction

In recent years, the rapid growth of artificial intelligence and deep learning has made it possible to generate highly realistic synthetic media, commonly known as deepfakes. These manipulated videos can alter a person’s identity, expressions, or speech in a way that appears natural to human observers. While such technology has useful applications in areas like entertainment, virtual reality, and content creation, it also introduces serious risks. Deepfakes can be misused for spreading misinformation, impersonation, cybercrime, and manipulation of public opinion, making it increasingly difficult to trust digital content.

Facial manipulation techniques such as identity swapping and expression transfer have become more advanced and accessible. As a result, distinguishing between real and manipulated videos has become a challenging problem, even for trained professionals. The difficulty lies in the fact that modern deepfake generation methods produce only very subtle artifacts, which are often not visible in the raw visual appearance of the video. This creates a strong need for automated systems that can reliably detect such manipulations.

Traditional approaches for deepfake detection primarily rely on spatial features extracted from images or video frames. These methods focus on inconsistencies in textures, facial structures, or pixel-level details. However, as generation techniques improve, these visible artifacts become less prominent, reducing the effectiveness of purely spatial methods. Moreover, many existing models struggle to capture complex relationships across different regions of the face, limiting their ability to detect more sophisticated manipulations.

To address these challenges, this project proposes a detection pipeline that combines frequency-based feature extraction with transformer-based learning. Instead of relying only on pixel-level information, the approach analyzes hidden patterns in the frequency domain, which often reveal inconsistencies introduced during the manipulation process. These features are then processed using a Vision Transformer model, which is capable of capturing both local and global dependencies within an image, improving detection performance.

In addition to classification, this work extends the functionality of the system by incorporating frame-level localization of manipulated regions. Rather than simply predicting whether a video is real or fake, the system identifies and highlights specific areas within each frame where manipulation is likely present. This enhancement makes the model more interpretable and useful for real-world applications, where understanding the location of forgery is as important as detecting it.



Furthermore, the proposed pipeline is evaluated not only on standard datasets but also on custom-generated videos to ensure practical relevance. This helps in assessing how well the model performs under realistic conditions, including variations in quality, lighting, and compression. Overall, the goal of this project is to develop a deepfake detection system that is accurate, robust, and interpretable, making it suitable for real-world deployment.

## 1.1 Motivation

The primary motivation for this work arises from the growing difficulty in reliably detecting deepfake content using existing approaches. As deepfake generation techniques continue to improve, the visual quality of manipulated videos has reached a level where traditional detection methods struggle to differentiate between real and fake content. Many existing systems rely heavily on surface-level visual cues, which are becoming increasingly insufficient because modern forgeries are designed to minimize visible artifacts. This gap highlights the need for a more robust and fundamentally different approach to deepfake detection.

Another key motivation is the limitation of relying solely on spatial features. While pixel-level and texture-based analysis can capture obvious inconsistencies, they often fail to detect subtle manipulations that are embedded in the underlying structure of the image. We recognized that important forensic information is often hidden in the frequency domain, where manipulation processes leave distinct but less visible traces. This led us to design our pipeline to explicitly extract and utilize frequency-based features, allowing the model to detect patterns that are not apparent in the raw visual representation.

We were also motivated by the need to improve how models understand relationships within an image. Conventional convolutional networks process information locally and require deep architectures to capture global context, which increases complexity without guaranteeing better understanding of long-range dependencies. To overcome this, we incorporated a transformer-based architecture in our pipeline, enabling the model to analyze the entire image holistically. This allows better detection of inconsistencies across different facial regions, which is critical in identifying sophisticated manipulations.

A major practical limitation we identified in existing systems is their lack of interpretability. Most deepfake detection models provide only a binary output, indicating whether a video is real or fake, without offering any explanation. This is insufficient for real-world applications such as digital forensics, content verification, or legal analysis, where understanding where the manipulation occurs is essential. To address this, we extended our pipeline beyond classification by introducing frame-level localization, where manipulated regions are highlighted using bounding boxes. This makes the system more transparent and useful for end users.

Finally, we were motivated by the need to build a system that performs well beyond

controlled experimental conditions. Many models show strong performance on benchmark datasets but fail when applied to real-world data due to variations in lighting, compression, and noise. To ensure practical relevance, we designed and tested our pipeline not only on standard datasets but also on our own generated videos. This helped us evaluate the robustness and adaptability of our approach in more realistic scenarios.

Overall, the motivation behind this work is to develop a deepfake detection pipeline that is not only accurate but also robust, interpretable, and practically applicable, by integrating frequency-domain analysis, transformer-based learning, and frame-level localization into a unified system designed entirely by us.

## 1.2 Literature Review

Deepfake detection has evolved through multiple research directions, ranging from traditional feature-based techniques to advanced deep learning and transformer-based approaches. Each category addresses specific challenges in detecting manipulated content, but also comes with its own limitations. This section presents a detailed review of key approaches relevant to video forgery detection.

### 1.2.1 Review of Existing Deepfake Detection Approaches

#### (a) CNN-Based Deepfake Detection — Chollet [8]

**Overview.** Convolutional Neural Networks (CNNs) have been one of the earliest and most widely adopted approaches for deepfake detection. These models are designed to learn spatial hierarchies of features directly from images, making them suitable for identifying visual inconsistencies introduced during manipulation. CNN-based methods focus on detecting artifacts such as unnatural textures, boundary mismatches, color inconsistencies, and blending errors around facial regions. As deepfake generation methods initially produced noticeable distortions, CNNs were highly effective in identifying such irregularities.

**Methodology.** In typical implementations, video data is first decomposed into individual frames. These frames are then preprocessed through face detection and alignment to isolate facial regions. The processed frames are passed through CNN architectures like Xception, ResNet, or EfficientNet to extract feature representations. These features are subsequently fed into classification layers to determine whether the input is real or manipulated. Some approaches further enhance performance by aggregating predictions across multiple frames to produce a final video-level decision.

#### **Key Findings.**

- CNNs can effectively learn discriminative spatial features without manual feature engineering.

- They perform well on benchmark datasets, especially when manipulation artifacts are visible.
- Pretrained CNN models enable transfer learning, reducing training time and improving accuracy.

#### **Limitations.**

- CNNs primarily focus on local patterns and may fail to capture global inconsistencies across the face.
- Their performance degrades when deepfake techniques produce high-quality, artifact-free outputs.
- These models act as black boxes, offering limited interpretability regarding decision-making.

### **(b) RNN and Temporal-Based Approaches — Guéra & Delp [9]**

**Overview.** While CNNs focus on spatial features, temporal-based approaches aim to capture inconsistencies across consecutive frames in a video. Deepfake videos often fail to perfectly replicate natural temporal dynamics such as eye blinking, facial muscle movements, and motion continuity. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, are used to model these sequential dependencies.

**Methodology.** In these approaches, spatial features are first extracted from individual frames using CNNs. These features are then passed as sequential inputs to RNN architectures like LSTM or GRU. The recurrent layers analyze temporal relationships between frames, identifying inconsistencies that may indicate manipulation. Some methods also incorporate optical flow or motion-based features to strengthen temporal analysis.

#### **Key Findings.**

- Temporal modeling improves detection performance in videos where spatial artifacts are minimal.
- RNN-based systems can capture subtle motion inconsistencies that are difficult to detect in single frames.
- Combining spatial and temporal features leads to more robust detection.

#### **Limitations.**

- High computational cost due to sequential data processing.

- Requires large amounts of labeled video data for effective training.
- Temporal inconsistencies may not always exist in well-generated deepfakes, reducing reliability.

### **(c) Optical Flow-Based Detection — Saikia et al [11]**

**Overview.** Optical flow-based methods analyze motion patterns between consecutive frames to detect irregularities caused by manipulation. Since deepfake generation may disrupt natural motion consistency, analyzing pixel movement provides an additional dimension for detection beyond static image analysis.

**Methodology.** Optical flow algorithms compute motion vectors that represent the displacement of pixels between frames. These motion maps are then used as input features for machine learning or deep learning models. Some approaches integrate optical flow with CNN or hybrid architectures to jointly analyze spatial and motion-based information.

#### **Key Findings.**

- Optical flow captures temporal inconsistencies that are not visible in static frames.
- Useful for detecting manipulations involving facial expressions and movements.
- Provides complementary information to spatial features.

#### **Limitations.**

- Computationally expensive, especially for high-resolution videos.
- Sensitive to noise, compression artifacts, and camera motion.
- Less effective in videos with minimal movement or stable scenes.

### **(d) Frequency-Domain Based Detection — Dosovitskiy et al [12]**

**Overview.** Frequency-domain approaches focus on identifying artifacts introduced during the generation process that are not easily visible in the spatial domain. Deepfake generation often disturbs the natural frequency distribution of images due to operations like upsampling, blending, and warping. These disturbances can be detected by analyzing transformed representations of images.

**Methodology.** Images are converted into frequency representations using techniques such as Discrete Cosine Transform (DCT), Fourier Transform, or wavelet transforms. These transformations decompose the image into different frequency bands, allowing the model to analyze fine-grained details. Features extracted from these domains are then used as inputs to machine learning or deep learning classifiers.

**Key Findings.**

- Frequency-based features reveal hidden artifacts missed by spatial analysis.
- Effective in detecting subtle inconsistencies in high-quality deepfakes.
- Enhances robustness when combined with spatial features.

**Limitations.**

- Frequency features alone may not capture contextual relationships across the image.
- Requires careful preprocessing and feature extraction design.
- Increased computational complexity compared to basic spatial methods.

**(e) Transformer-Based Deepfake Detection — Zhao et al [13]**

**Overview.** Transformer-based models have emerged as a powerful alternative to CNNs for visual analysis tasks. These models use self-attention mechanisms to capture relationships between different regions of an image, enabling a more holistic understanding of visual content. In deepfake detection, transformers are used to identify inconsistencies across the entire face rather than focusing on local regions.

**Methodology.** Images are divided into fixed-size patches, which are then converted into embeddings. These embeddings are processed through multiple transformer layers, where self-attention mechanisms learn interactions between patches. A classification token aggregates global information and is used to predict whether the input is real or fake. Attention maps can also be analyzed to understand which regions influenced the decision.

**Key Findings.**

- Captures long-range dependencies more effectively than CNNs.
- Provides global context, improving detection of subtle manipulations.
- Offers partial interpretability through attention visualization.

**Limitations.**

- Requires large-scale datasets for effective training.
- Computationally intensive compared to traditional models.
- Sensitive to hyperparameter tuning and input representation.

### 1.2.2 Synthesis of Literature and Identified Research Gaps

Across the reviewed studies, several consistent limitations emerge:

The reviewed studies collectively demonstrate that deepfake detection has evolved through diverse methodological directions, including spatial feature learning, temporal modeling, motion analysis, frequency-domain techniques, and transformer-based architectures. Each category contributes to improving detection performance, yet they address only specific aspects of the problem. Spatial-based approaches, particularly CNN models, are effective in identifying visible artifacts such as texture inconsistencies and blending errors. However, as deepfake generation techniques advance, these artifacts become increasingly subtle, reducing the effectiveness of purely spatial methods.

Temporal-based approaches extend detection by analyzing inconsistencies across video frames, capturing unnatural motion patterns and behavioral anomalies. While these methods enhance detection in dynamic scenarios, their performance is highly dependent on the presence of detectable temporal irregularities, which are often minimized in high-quality manipulations. Similarly, optical flow-based techniques provide motion-related insights but suffer from high computational complexity and sensitivity to noise, limiting their scalability in real-world applications.

Frequency-domain methods introduce a different perspective by focusing on hidden artifacts embedded in transformed representations of images. These approaches reveal subtle inconsistencies that are not observable in the spatial domain, making them valuable for detecting refined manipulations. However, they often lack the ability to capture global contextual relationships across the face, which are important for understanding complex forgeries. Transformer-based models address this limitation by enabling global attention and capturing long-range dependencies across image regions. Despite their advantages, these models require large-scale data and significant computational resources, making them less accessible for lightweight implementations.

Overall, the synthesis of these studies highlights that while each approach improves detection in specific scenarios, no single method is sufficient to handle the wide variety of deepfake manipulations. This indicates the necessity of combining complementary techniques to achieve a more robust and generalized detection system.

## 1.3 Existing Challenges

Despite significant progress in deepfake detection, several challenges continue to limit the effectiveness and real-world applicability of current approaches. One of the primary challenges is the rapid evolution of deepfake generation techniques, which continuously reduce visible artifacts and improve temporal consistency. As a result, many detection models struggle to generalize across different types of manipulations, especially when trained on specific datasets [1].

Another major challenge is the trade-off between accuracy and computational efficiency. High-performing models, particularly those based on deep neural networks and transformer architectures, often require extensive computational resources, including high-end GPUs and large memory capacity. This makes them unsuitable for deployment in resource-constrained environments such as edge devices or real-time applications [13].

Generalization across varying video qualities is also a critical issue. In real-world scenarios, videos are often compressed, noisy, or captured under different lighting conditions. Many detection models show strong performance on benchmark datasets but fail to maintain accuracy when applied to such real-world data. This highlights the need for models that are robust to compression artifacts and environmental variations [7].

Another limitation is the lack of interpretability in most detection systems. Many models provide only a binary classification output without indicating the specific regions responsible for manipulation. This lack of transparency reduces user trust and limits the applicability of these systems in domains such as digital forensics and legal investigations, where explainability is essential [5].

Additionally, preprocessing requirements in many existing approaches add to system complexity. Techniques involving face detection, alignment, temporal sequencing, or motion estimation increase processing time and resource consumption. This makes it challenging to design efficient pipelines that balance performance and practicality [4].

In summary, current deepfake detection methods face challenges related to generalization, computational cost, interpretability, and robustness under real-world conditions. Addressing these issues is essential for developing reliable and scalable solutions that can effectively combat the growing threat of deepfake media [7].

## 1.4 Objectives of the Project

### 1.4.1 Develop a Robust Deepfake Detection Pipeline

A key objective of this project is to design and implement a complete deepfake detection pipeline capable of processing real-world video data efficiently. The pipeline

includes stages such as frame extraction, face detection, preprocessing, feature transformation, and classification. Special emphasis is placed on ensuring consistency across frames by handling variations in lighting, pose, and video quality. A well-structured pipeline is essential because the reliability of each stage directly impacts the overall detection performance.

#### **1.4.2 Leverage Frequency-Domain Features for Improved Detection**

The project aims to incorporate frequency-domain analysis to capture subtle artifacts introduced during manipulation. By applying transformations such as the Stationary Wavelet Transform, the system extracts hidden patterns that are not visible in the spatial domain. This objective focuses on enhancing the model's ability to detect high-quality deep-fakes where traditional pixel-based methods fail. Utilizing frequency features improves robustness and enables better identification of fine-grained inconsistencies.

#### **1.4.3 Utilize Transformer-Based Architecture for Global Feature Learning**

Another objective is to integrate a transformer-based model to overcome the limitations of traditional convolutional networks. The transformer architecture allows the system to analyze relationships across the entire image using self-attention mechanisms. This enables better understanding of global dependencies between facial regions, which is critical for detecting complex manipulations. The goal is to achieve higher accuracy by combining local feature extraction with global contextual awareness.

#### **1.4.4 Enable Frame-Level Localization of Manipulated Regions**

In addition to classification, the project aims to provide interpretability by identifying and highlighting manipulated regions within each frame. This is achieved by implementing frame-level localization using bounding boxes around suspicious areas. The objective is to move beyond simple binary predictions and offer visual explanations that indicate where manipulation has occurred. This makes the system more useful for practical applications such as forensic analysis and content verification.

#### **1.4.5 Ensure Robustness and Practical Applicability**

The project also focuses on evaluating the system under realistic conditions. This includes testing on both benchmark datasets and custom-generated videos to assess performance across different scenarios. The objective is to ensure that the model generalizes well to variations in compression, noise, and lighting conditions. Additionally, efforts are made to balance accuracy with computational efficiency, making the system suitable for practical



deployment.

## **1.5 Organisation of the Report**

The remainder of this report is structured to systematically present the development and evaluation of the proposed deepfake detection system.

Chapter 2 discusses the background concepts and related methodologies relevant to the project. It covers fundamental aspects of video forgery detection, frequency-domain analysis, and transformer-based learning, which form the theoretical foundation of the proposed approach.

Chapter 3 describes the proposed methodology and system architecture in detail. It explains the complete pipeline, including video frame extraction, face detection, preprocessing, frequency feature extraction, and classification using the transformer model. It also includes the implementation of frame-level localization for identifying manipulated regions.

Chapter 4 presents the dataset and experimental setup. It includes details of the datasets used, preprocessing steps, training configurations, and evaluation metrics employed to assess the performance of the model.

Chapter 5 discusses the results and analysis of the proposed system. It provides a detailed evaluation of model performance, comparisons with baseline approaches, and analysis of the effectiveness of frame-level localization.

# Chapter 2 Foundational Study

## Introduction

Deepfake technology has rapidly evolved into a powerful tool capable of generating highly realistic manipulated media, making it increasingly difficult to distinguish between authentic and forged content. This advancement has introduced serious concerns across multiple domains, including digital security, media authenticity, and information integrity. As manipulated videos become more sophisticated, there is a growing need to understand the underlying techniques used for both generation and detection of such content.

This chapter provides the foundational knowledge required for understanding the proposed deepfake detection system. It begins with an overview of video forgery techniques and the challenges associated with identifying manipulated content. It then explores different feature extraction approaches, particularly focusing on the importance of frequency-domain analysis in revealing hidden artifacts that are not easily visible in raw images.

In addition, this chapter discusses modern deep learning approaches used in visual analysis, with special emphasis on transformer-based architectures. These models have gained significant attention due to their ability to capture global relationships within images, making them highly effective for complex tasks such as forgery detection.

The concepts covered in this chapter form the theoretical backbone of the project. A clear understanding of these topics is essential for interpreting the design choices and methodology adopted in the proposed system. By combining insights from video analysis, frequency transformations, and advanced learning models, this chapter sets the stage for the development of a robust and interpretable deepfake detection pipeline.

## 2.1 Background Study

### 2.1.1 Fundamentals of Video Forgery Detection

#### 2.1.1.1 Video forgery detection

Video forgery detection is a critical area in digital forensics that focuses on identifying manipulated or synthetically generated video content. With the advancement of deep learning and generative models, video manipulations such as identity swapping, expression reenactment, and full synthetic face generation have become increasingly realistic. These manipulations often preserve visual continuity and temporal coherence, making manual

detection nearly impossible and necessitating automated solutions.

Early approaches to video forgery detection relied on handcrafted features derived from visual and temporal inconsistencies. These methods analyzed artifacts such as abnormal textures, color mismatches, edge discontinuities, and irregular lighting conditions. Temporal inconsistencies, including unnatural blinking patterns, lip synchronization errors, and motion discontinuities across frames, were also used as indicators of manipulation. While these techniques were effective against simpler forgeries, they struggle significantly with modern deepfake methods that are designed to minimize such visible artifacts.

With the rise of deep learning, detection approaches have shifted toward data-driven models capable of automatically learning discriminative features from large datasets. Convolutional Neural Networks (CNNs) are commonly used to extract spatial features from individual frames, while hybrid models combine CNNs with temporal networks to capture motion-based inconsistencies. These approaches have shown significant improvements in detection accuracy, particularly when trained on large-scale datasets such as FaceForensics++, which provides diverse examples of manipulated and real videos under different compression levels.

Despite their success, deep learning-based methods face several challenges. Many models act as black boxes, offering limited interpretability and making it difficult to understand the reasoning behind predictions. Additionally, their performance is often sensitive to variations in video quality, compression, and frame-level noise. These limitations highlight the need for more robust and explainable detection systems that can generalize well across different real-world scenarios.[1]

#### **2.1.1.2 Frame-Level Analysis and Localization in Video Processing**

Frame-level analysis is an important aspect of modern deepfake detection systems, especially when moving beyond simple classification tasks. Instead of treating an entire video as a single entity, frame-level processing allows the model to analyze each frame independently, making it possible to detect localized inconsistencies and variations over time. Localization techniques aim to identify specific regions within a frame where manipulation has occurred, typically through bounding boxes or heatmaps that highlight suspicious areas. In deepfake detection, manipulated regions often correspond to facial components such as eyes, lips, or cheeks, where blending and synthesis artifacts are more prominent. Such localization can be implemented using feature activation maps, attention mechanisms, or post-processing techniques applied to model outputs — transforming the system from a black-box classifier into a more transparent tool that provides visual evidence of manipulation. This interpretability is particularly important in applications like digital forensics, where understanding the location of tampering is as crucial as detecting it.

Frame-level analysis also enables better handling of temporal inconsistencies. Even

if some frames appear authentic, manipulated frames within the sequence can still be detected, improving overall robustness. This makes the approach suitable for real-world scenarios where manipulation may not be uniform across all frames [5].

## **2.1.2 Feature Extraction and Preprocessing Techniques**

### **2.1.2.1 Frequency-Domain Feature Analysis**

Frequency-domain analysis has emerged as a powerful technique for detecting deepfake content by revealing artifacts that are not easily observable in the spatial domain. Manipulation processes such as blending, warping, upsampling, and compression introduce subtle distortions in the frequency distribution of images. These distortions may not be visible to the human eye but can be effectively captured through frequency transformations.

Unlike spatial analysis, which focuses on pixel intensities, frequency-domain methods analyze how image information is distributed across different frequency bands. High-frequency components typically represent fine details such as edges and textures, while low-frequency components capture general structure and illumination. Deepfake generation often disrupts this natural distribution, resulting in detectable anomalies in specific frequency ranges.

Wavelet-based techniques are particularly effective for this type of analysis. They decompose an image into multiple subbands, allowing simultaneous localization in both spatial and frequency domains. This is especially useful for identifying localized artifacts introduced during manipulation. Among these techniques, the Stationary Wavelet Transform (SWT) offers significant advantages. Unlike traditional wavelet transforms, SWT does not involve downsampling, preserving the original spatial resolution of the image. This property ensures that fine-grained details are retained, making it highly suitable for detecting subtle inconsistencies at specific regions of the face.

By analyzing different frequency subbands, it becomes possible to highlight irregular patterns caused by synthetic generation processes. These features can then be used as inputs to machine learning or deep learning models, enhancing their ability to detect sophisticated forgeries. However, frequency-domain analysis alone may not fully capture the contextual relationships within an image, which necessitates its integration with advanced learning architectures.[2]

### **2.1.2.2 Face Detection and Preprocessing Techniques**

Face detection and preprocessing form a critical foundation in any deepfake detection pipeline. Since manipulations are primarily focused on facial regions, isolating the face

accurately from each video frame significantly improves the quality of feature extraction and subsequent classification. Raw video frames often contain background noise, varying lighting conditions, occlusions, and multiple faces, which can negatively impact model performance if not handled properly.

Modern face detection techniques rely on deep learning-based models such as Multi-task Cascaded Convolutional Networks (MTCNN), RetinaFace, and Haar Cascade variants. Among these, MTCNN is widely used due to its ability to detect faces along with key facial landmarks such as eyes, nose, and mouth. These landmarks enable precise alignment and cropping of facial regions, ensuring consistency across frames. This consistency is essential because variations in pose, scale, or orientation can introduce unwanted noise into the learning process.

Preprocessing steps typically include resizing images to a fixed resolution, normalization of pixel values, and removal of irrelevant background information. In video-based pipelines, frame extraction strategies (such as sampling at fixed frame rates) are also applied to reduce redundancy and computational load. Proper preprocessing ensures that the model focuses only on relevant facial features, thereby improving both efficiency and accuracy of the detection system. [4].

### **2.1.2.3 Video Processing and Frame Extraction Techniques**

Efficient video processing is essential for handling large-scale datasets and real-time applications. Since processing every frame can be computationally expensive and often redundant, frame extraction strategies are employed to balance computational efficiency with information retention. Common approaches include uniform frame sampling, where frames are extracted at fixed intervals, and keyframe extraction, where only significant frames are selected based on motion or scene changes. In deepfake detection pipelines, selecting representative frames reduces training time and memory usage without significantly affecting performance.

Video decoding and processing are typically performed using libraries such as OpenCV or FFmpeg, which allow efficient extraction, resizing, and manipulation of frames. Preprocessing steps such as color space conversion, normalization, and augmentation are additionally applied to improve model generalization. Proper frame extraction is especially important when integrating with deep learning models, as it ensures that the input data is both manageable and informative while maintaining consistency across different videos — a property crucial for training robust detection systems [6].

## **2.1.3 Deep Learning Models and Integrated Detection Approaches**

### **2.1.3.1 Vision Transformers in Visual Analysis**

Vision Transformers (ViTs) represent a significant advancement in computer vision by introducing transformer-based architectures for image and video analysis. Unlike traditional convolutional neural networks, which process images using local receptive fields, ViTs analyze images globally by modeling relationships between different regions through self-attention mechanisms. The core idea is to divide an image into fixed-size patches and treat each patch as a token, analogous to words in natural language processing. These tokens are embedded into a feature space and processed through multiple transformer layers, where self-attention allows each patch to interact with every other patch in the image — enabling the model to capture long-range dependencies and global contextual information in a single stage.

This global perspective is particularly beneficial for deepfake detection, where inconsistencies may not be confined to a single region but distributed across multiple facial areas. Mismatches between facial expressions, lighting conditions, or texture patterns across different parts of the face can be effectively captured using attention mechanisms. Additionally, the attention maps generated by transformers provide a degree of interpretability by indicating which regions of the image contributed most to the final decision.

Despite their advantages, Vision Transformers typically require large amounts of training data and are computationally more intensive than traditional CNN-based models. However, when combined with appropriate feature representations — such as those derived from frequency-domain analysis — they can significantly enhance detection performance, making them well-suited for tasks like deepfake detection where both local details and global context are important [3].

### **2.1.3.2 Integration of Multi-Domain Features in Detection Pipelines**

Modern deepfake detection systems increasingly rely on the integration of multiple feature domains to improve robustness. Instead of depending solely on spatial, temporal, or frequency features, combining these domains allows the model to capture complementary information, leading to better performance across diverse scenarios. In a typical integrated pipeline, spatial features capture visible inconsistencies, frequency features reveal hidden artifacts, and transformer-based architectures model global relationships. Preprocessing ensures that inputs are standardized, while frame-level analysis enables localization and interpretability. This multi-stage approach addresses the limitations of individual methods and creates a more comprehensive detection framework.

Feature fusion can be performed at different stages — such as early fusion, which

combines raw features before processing, or late fusion, which combines outputs from different models. The choice of fusion strategy depends on the specific design of the pipeline and computational constraints. This integration is particularly important for handling real-world data, where variations in quality, compression, and environmental conditions can affect detection performance. By leveraging diverse feature representations, the system becomes more adaptable and resilient to different types of manipulations [7].

## Chapter Summary

This chapter presented the foundational concepts required to understand and develop an effective deepfake detection system. It began with an overview of video forgery detection, highlighting how manipulation techniques have evolved from simple edits to highly realistic synthetic content, making detection increasingly challenging [1]. The discussion emphasized the transition from traditional handcrafted feature-based methods to modern deep learning approaches, along with their respective strengths and limitations.

The chapter then explored frequency-domain feature analysis, explaining how manipulation introduces subtle artifacts that are not easily visible in the spatial domain. Techniques such as wavelet transforms, particularly the Stationary Wavelet Transform, were discussed for their ability to capture fine-grained inconsistencies while preserving spatial resolution [2]. This provided a strong foundation for understanding the importance of frequency-based features in detecting sophisticated forgeries.

Further, the role of Vision Transformers in visual analysis was examined, focusing on their ability to capture global dependencies using self-attention mechanisms [3]. Unlike conventional convolutional models, transformers provide a holistic understanding of images, making them well-suited for identifying inconsistencies across different facial regions.

Additional components of the detection pipeline were also discussed, including face detection and preprocessing using methods such as MTCNN [4], frame-level analysis and localization techniques supported by visualization approaches like Grad-CAM [5], and efficient video processing techniques using tools such as OpenCV [6]. These elements are crucial for ensuring consistent input data, reducing computational overhead, and improving interpretability through region-based highlighting of manipulated areas.

Overall, this chapter established the theoretical and technical groundwork for the proposed system by integrating insights from multiple domains [7]. The combination of spatial processing, frequency-domain analysis, and transformer-based learning forms the basis for a robust, accurate, and interpretable deepfake detection pipeline, which is further detailed in the subsequent chapters.

# Chapter 3 System Design and Methodology

## 3.1 Introduction

The proliferation of synthetically manipulated video content — colloquially referred to as *deepfakes* — has emerged as one of the most significant technological threats of the twenty-first century. Enabled by advances in generative adversarial networks (GANs), variational autoencoders (VAEs), and identity-swapping algorithms, deepfake videos are capable of substituting one person’s likeness onto another’s body with photorealistic fidelity, or transferring the facial expressions and mouth movements of a target speaker onto a source subject. The consequences of such manipulations span a wide and alarming spectrum: from the non-consensual fabrication of intimate imagery, to the orchestration of large-scale political disinformation campaigns, to fraud in financial and legal domains where video evidence carries evidentiary weight.

Existing detection approaches have historically treated the deepfake detection problem as a binary classification task — producing a single verdict of *real* or *fake* at the video level. While such methods offer computational simplicity, they suffer from two fundamental limitations that severely curtail their practical utility. First, they provide no spatial interpretability: a verdict of “fake” does not reveal *which region* of the face was manipulated, nor *how* the manipulation was performed. Second, video-level labels obtained by simple majority voting across frame predictions are brittle to adversarial perturbations or low-quality compression, both of which are ubiquitous in real-world distribution conditions.

This chapter presents a comprehensive account of the system designed and implemented to address these limitations. The proposed pipeline — hereafter referred to as **SWTVIT** — integrates the Stationary Wavelet Transform (SWT) as a preprocessing front-end with a fine-tuned Vision Transformer (ViT) backbone to perform four-class classification across the manipulation categories defined by the FaceForensics++ benchmark. Beyond classification, the system incorporates a GradCAM-based spatial attention module that produces frame-wise localization maps, enabling the precise identification of the tampered facial region within each extracted frame. Annotated frames are subsequently reassembled into a coherent output video, yielding a temporally consistent, interpretable, and visually compelling deepfake detection artefact suitable for both forensic analysis and real-world deployment.

The chapter is organized as follows. Section 3.2 provides a holistic overview of the system architecture and the rationale behind each design decision. Section 3.3 details the individual components of the neural architecture, including the wavelet preprocessing



stage, the ViT backbone, and the classification head. Section 3.4 describes the end-to-end operational workflow, encompassing data ingestion, preprocessing, training, inference, and output generation. Section 3.5 synthesizes the key contributions and transitions to the experimental evaluation presented in subsequent chapters.

## 3.2 System Design

### 3.2.1 Design Philosophy and Motivation

The design of SWTVIT is guided by three overarching principles that collectively distinguish it from prior work in the deepfake detection literature.

**Principle 1 — Frequency-domain preprocessing.** The pixel-domain representations of deepfake frames are, by construction, designed to be perceptually indistinguishable from genuine footage. Autoencoder-based face-swapping methods such as *Deepfakes* and *FaceSwap* generate reconstructed faces whose low-frequency content faithfully replicates the source identity while introducing characteristic blending artefacts at high spatial frequencies, particularly along the boundary between the synthetic face region and the native background. A detector that operates exclusively in the pixel domain may fail to exploit these high-frequency discriminative cues, especially when the input has been subjected to lossy compression. The Stationary Wavelet Transform was therefore selected as a pre-processing stage that explicitly decomposes each face image into its constituent frequency subbands, surfacing the otherwise latent artefact signatures that constitute the most reliable discriminators of manipulation.

**Principle 2 — Global context through self-attention.** Convolutional neural networks, despite their documented efficacy on natural image classification tasks, are inherently local in their receptive field growth. Detecting the subtle spatial inconsistencies introduced by deepfake generation — such as asymmetric lighting, unnatural eye blinking patterns, or discontinuous skin textures across the face boundary — requires the ability to reason simultaneously over spatially distant regions of the face. Vision Transformers achieve this through their multi-head self-attention mechanism, which computes pairwise relationships between all patch tokens in a single layer, without the need to stack many convolutional layers to achieve a globally receptive field. This makes ViT architecturally well-suited to the deepfake detection problem.

**Principle 3 — Spatial interpretability through localization.** A detection system that merely classifies a video as manipulated, without specifying the spatial locus of manipulation, provides insufficient information for forensic investigators or content moderation systems. SWTVIT therefore incorporates a GradCAM-based localization module that generates a spatial attention map for each processed frame, which is subsequently thresholded and converted into a bounding box delineating the tampered region. This design choice

transforms the system from a black-box classifier into an interpretable forensic instrument.

### 3.2.2 Dataset: FaceForensics++

All model training, validation, and testing was conducted on the **FaceForensics++** (FF++) benchmark dataset [1], the de facto standard for deepfake detection research. FF++ comprises 1000 original YouTube videos paired with their corresponding manipulated counterparts generated by four distinct forgery methods:

- **Deepfakes (DF):** An autoencoder-based identity-swap method that trains a shared encoder and two identity-specific decoders to transfer one person’s face onto another’s head pose and expression. This method produces full face replacements and is characterised by boundary blending artefacts.
- **FaceSwap (FS):** A computer-graphics-based face replacement pipeline that fits a 3D morphable model to both source and target faces and projects the source texture onto the target geometry. Unlike the autoencoder-based Deepfakes method, FaceSwap does not alter facial expressions and produces somewhat more geometrically consistent swaps, though at the expense of photorealistic skin texture fidelity.
- **Face2Face (F2F):** A real-time facial reenactment system that transfers the expressions — particularly mouth and eye movements — from a source actor to a target subject, while preserving the target’s appearance and identity. This method introduces characteristic artefacts in the mouth interior region.
- **NeuralTextures (NT):** A neural rendering approach that learns a texture map of the target face and synthesizes novel renderings of the target under the driving pose and expression of the source, producing high-fidelity results but with distinctive rendering artefacts in fine-grained facial structures.

For the purposes of this project, the three manipulation categories directly relevant to the autoencoder and expression-reenactment paradigms — namely *Deepfakes*, *FaceSwap*, and *Face2Face* — were selected alongside the *Original* (genuine) class, constituting a **four-class classification problem**. All videos were downloaded in the **c23** compression format (H.264 with a constant rate factor of 23), which represents a balance between file size and perceptual quality, and is the most commonly used format in real-world deepfake distribution scenarios. A total of **125 videos per class** were utilized, yielding 500 videos in aggregate.

### 3.2.3 Data Partitioning Strategy

A critical design decision concerns the granularity at which the dataset is split into training, validation, and test subsets. A naive frame-level split — in which individual

frames are randomly assigned to splits — introduces **frame-level data leakage**: frames extracted from the same video may appear in both the training and test sets, causing the model to memorize video-specific visual patterns rather than learning generalizable manipulation artefacts. This leads to artificially inflated test accuracy that does not reflect genuine out-of-distribution generalization.

SWTVIT employs a strict **video-level stratified split**, in which all frames extracted from a given video are constrained to appear exclusively within a single partition. The split ratios were set to 70% training, 15% validation, and 15% test, applied independently within each class to preserve class balance across all three partitions. With 125 videos per class, this yields approximately:

- **Training set:** 87 videos per class  $\times$  4 classes = 348 videos total
- **Validation set:** 19 videos per class  $\times$  4 classes = 76 videos total
- **Test set:** 19 videos per class  $\times$  4 classes = 76 videos total

The split is implemented with a fixed random seed (42) to ensure full reproducibility across experimental runs. This partitioning strategy guarantees that model performance estimates on the test set constitute a faithful measure of generalization to unseen video content.

### 3.2.4 Frame Extraction

Raw video files were processed using **FFmpeg**, an industry-standard multimedia processing framework, to extract individual frames suitable for downstream face detection and feature extraction. The extraction rate was set to **1 frame per second** (fps = 1), with a maximum cap of **50 frames per video**. This configuration provides adequate temporal coverage while remaining computationally tractable within the resource constraints of the experimental environment (Google Colab with NVIDIA Tesla T4 GPU).

The extraction pipeline implements an efficient caching mechanism: if a video’s output directory already exists and is populated with PNG files from a previous extraction run, the extraction step is bypassed and the cached frames are loaded directly. This design choice is critical for iterative experimental workflows in which the model or preprocessing parameters are modified without altering the underlying video data.

Frames exhibiting FFmpeg decoding errors — typically arising from corrupted video containers or malformed bitstreams — are handled gracefully by detecting non-zero return codes and returning an empty frame list, causing the affected video to be silently excluded from downstream processing without disrupting the pipeline.

### 3.2.5 Face Detection and Alignment

Deepfake manipulation acts specifically upon the facial region of a video; consequently, restricting the model’s input to a tightly cropped, geometrically normalized face patch — rather than the full video frame — substantially reduces the feature space to the region of forensic interest. SWTVIT employs the **Multi-task Cascaded Convolutional Network (MTCNN)** [4] for face detection and landmark localization, implemented via the *facenet-pytorch* library.

MTCNN operates as a three-stage cascaded detector: a shallow Proposal Network (P-Net) generates candidate face regions via a sliding window approach; a Refine Network (R-Net) filters false positives using more discriminative features; and an Output Network (O-Net) produces refined bounding boxes and five facial landmark coordinates (two eyes, nose tip, and two mouth corners). The detector is configured with:

- **Minimum face size:** 30 pixels — sufficient to detect faces in moderate-distance shots while excluding spurious small-scale detections
- **Confidence thresholds:** [0.5, 0.6, 0.6] for P-Net, R-Net, and O-Net respectively
- **Face selection policy:** When multiple faces are detected in a frame, the largest face by bounding box area is selected, on the assumption that the primary subject occupies the most prominent facial region

Detected faces are subjected to **geometric alignment** using the eye landmark coordinates to estimate a rotation angle that brings the inter-ocular axis to horizontal. An affine rotation matrix is applied to the face crop, followed by resizing to a standardized  $224 \times 224$  pixel resolution. This alignment step ensures that the model receives geometrically consistent face representations irrespective of head pose variation across frames or videos.

An important architectural decision concerns the placement of the MTCNN detector within the computational graph. Running MTCNN inside a *PyTorch DataLoader* worker process would require spawning CUDA contexts in child processes, which is known to cause non-deterministic deadlocks in the PyTorch multiprocessing backend. SWTVIT therefore performs all face detection in the **main process** prior to the training loop, saving detected face crops as PNG files to a dedicated cache directory. *DataLoader* workers subsequently operate exclusively on CPU-resident PNG files and perform only the lightweight SWT and tensor transformation steps, making the use of multiple parallel workers safe and performant.

### 3.2.6 Class Imbalance Mitigation

Even with stratified splitting, the effective number of training samples per class may vary as a consequence of heterogeneous face detection rates and video lengths. To pre-

vent the model from developing a bias towards the majority class, SWTVIT employs a **Weighted Random Sampler** during training. For each class  $c$ , a sample weight  $w_c = 1/n_c$  is computed, where  $n_c$  is the number of training samples belonging to class  $c$ . Each training sample is then assigned its corresponding class weight, and the DataLoader draws batches by sampling proportionally to these weights with replacement, ensuring that all four classes contribute approximately equally to each mini-batch regardless of their raw frequency in the training set.

### 3.3 Model Illustration

#### 3.3.1 Overview of the SWTVIT Architecture

The SWTVIT model comprises three sequentially composed modules: (1) a *Stationary Wavelet Transform* preprocessing block that converts each aligned face crop into a frequency-domain representation emphasizing high-frequency manipulation artefacts; (2) a *Vision Transformer backbone* that extracts rich hierarchical feature representations from the wavelet-processed input; and (3) a *classification head* that maps the extracted features to a probability distribution over the four target classes. The following subsections describe each module in detail.

#### 3.3.2 Stationary Wavelet Transform (SWT) Preprocessing

The core insight motivating the use of the Stationary Wavelet Transform as a preprocessing stage is that deepfake generation methods introduce characteristic high-frequency artefacts that are substantially more discriminative in the frequency domain than in the pixel domain. Autoencoder-based methods, in particular, produce blending boundaries that manifest as sharp transitions in the wavelet detail subbands.

Given an aligned face crop  $\mathbf{I} \in R^{224 \times 224 \times 3}$ , the SWT preprocessing proceeds as follows:

**Step 1 — Grayscale conversion.** The RGB image is converted to a single-channel grayscale image  $\mathbf{G} \in R^{224 \times 224}$  using the standard luminance formula:

$$\mathbf{G} = 0.299 \mathbf{I}_R + 0.587 \mathbf{I}_G + 0.114 \mathbf{I}_B \quad (3.1)$$

**Step 2 — Two-dimensional SWT decomposition.** A single-level 2D Stationary Wavelet Transform is applied to  $\mathbf{G}$  using the **Haar** mother wavelet. The Haar wavelet is selected for its compact support — it has the smallest possible support among all orthogonal wavelets — making it maximally sensitive to sharp local transitions such as blending

boundaries. The SWT decomposes  $\mathbf{G}$  into four subbands of identical spatial dimensions:

$$SWT(\mathbf{G}) = \{cA, cH, cV, cD\} \quad (3.2)$$

where  $cA$  is the approximation (low-frequency) subband, and  $cH$ ,  $cV$ ,  $cD$  are the horizontal, vertical, and diagonal detail (high-frequency) subbands respectively. Unlike the standard Discrete Wavelet Transform, the SWT does not apply downsampling between decomposition levels, preserving the full spatial resolution of each subband — a property that is critical for pixel-accurate artefact localization.

**Step 3 — High-frequency magnitude computation.** The approximation subband  $cA$  is discarded, as it encodes the low-frequency content of the face that carries identity information rather than manipulation artefacts. The three detail subbands are combined into a single high-frequency magnitude map  $\mathbf{C}$  according to the equation proposed in the SWTVIT paper:

$$\mathbf{C} = \sqrt{cH^2 + cV^2 + cD^2} \quad (3.3)$$

This combined magnitude captures artefacts that may manifest preferentially in any of the three directional subbands — horizontal blending seams, vertical boundary discontinuities, or diagonal GAN checkerboard patterns — without requiring the model to explicitly attend to each subband independently.

**Step 4 — Min-max normalization.** The magnitude map  $\mathbf{C}$  is normalized to the unit interval:

$$\hat{\mathbf{C}} = \frac{\mathbf{C} - \min(\mathbf{C})}{\max(\mathbf{C}) - \min(\mathbf{C}) + \epsilon} \quad (3.4)$$

where  $\epsilon = 10^{-8}$  is a small constant to prevent division by zero in the degenerate case of a uniform image.

**Step 5 — Channel replication.** Since the ViT backbone expects a three-channel input tensor of shape  $(3, 224, 224)$  consistent with its ImageNet pretraining, the normalized magnitude map  $\hat{\mathbf{C}} \in R^{224 \times 224}$  is replicated across three identical channels to yield the SWT feature tensor [2]:

$$\mathbf{F}_{SWT} = \text{stack}([\hat{\mathbf{C}}, \hat{\mathbf{C}}, \hat{\mathbf{C}}], \text{axis} = -1) \in R^{224 \times 224 \times 3} \quad (3.5)$$

This tensor is subsequently normalized using ImageNet channel statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) to ensure compatibility with the pre-trained ViT backbone’s input distribution.

### 3.3.3 Vision Transformer Backbone

The SWT feature tensor is processed by a **Vision Transformer** operating in the image classification configuration. Specifically, the `vit_small_patch16_224` architecture

from the *timm* library is employed, pretrained on ImageNet-21k. This model configuration offers a favorable trade-off between representational capacity and computational efficiency:

- **Total parameters:** 21,667,972 (approximately 21.7 million)
- **Embedding dimension:** 384
- **Number of attention heads:** 6
- **Number of transformer blocks:** 12
- **Patch size:**  $16 \times 16$  pixels
- **Number of patches:**  $(224/16)^2 = 196$  spatial patches + 1 [CLS] token = 197 tokens total

The ViT processes the input as follows. The  $224 \times 224$  input image is partitioned into a grid of non-overlapping  $16 \times 16$  patches. Each patch is linearly projected into an embedding vector of dimension 384 via a learned patch embedding layer. A learnable [CLS] token is prepended to the sequence of patch embeddings, and sinusoidal positional embeddings are added to each token to encode spatial position information. The resulting sequence of 197 tokens is processed by 12 successive Transformer encoder blocks, each comprising a Multi-Head Self-Attention (MHSA) layer and a position-wise Feed-Forward Network (FFN) with GELU activations, connected via residual connections and preceded by Layer Normalization.

The multi-head self-attention mechanism is central to the model’s capability. For each attention head  $h$ , the attention matrix is computed as [14]:

$$Attention(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = softmax\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}\right) \mathbf{V}_h \quad (3.6)$$

where  $\mathbf{Q}_h$ ,  $\mathbf{K}_h$ ,  $\mathbf{V}_h$  are the query, key, and value projections for head  $h$ , and  $d_k = 64$  is the per-head dimension. Critically, this operation computes pairwise attention weights between *all pairs* of the 197 tokens in a single layer, enabling the model to simultaneously attend to spatially distant patch pairs — for example, to detect inconsistencies between the forehead texture and the chin region that may arise from imperfect face blending.

After processing through all 12 Transformer blocks, the [CLS] token embedding (a vector of dimension 384) serves as the global representation of the input image, aggregating information from all 196 spatial patches through the cascaded self-attention layers. The [CLS] token is extracted via global pooling on the token dimension and passed to the classification head.

### 3.3.4 Classification Head

The classification head is a lightweight module appended to the ViT backbone to adapt the pretrained representations to the four-class deepfake detection task. It comprises three layers:

1. **Layer Normalization:** Applied to the 384-dimensional [CLS] embedding to stabilize the distribution of features entering the classification layers.
2. **Dropout:** A dropout layer with rate  $p = 0.3$  is applied during training to provide regularization and prevent overfitting to the relatively small training set. Dropout is disabled during inference.
3. **Linear projection:** A fully connected layer maps the 384-dimensional feature vector to a 4-dimensional output logit vector:

$$\hat{\mathbf{y}} = \mathbf{W} \cdot \text{LayerNorm}(\text{Dropout}(\mathbf{z}_{CLS})) + \mathbf{b} \quad (3.7)$$

where  $\mathbf{W} \in R^{4 \times 384}$  and  $\mathbf{b} \in R^4$  are learnable parameters.

The four output logits correspond to the classes {Real, Deepfake, FaceSwap, Face2Face} respectively. During training, these logits are passed directly to the cross-entropy loss function. During inference, a softmax function converts them to a probability distribution, and the predicted class is the argument of the maximum probability.

### 3.3.5 GradCAM-based Spatial Localization

To produce frame-wise localization of the tampered region, SWTVIT employs **Gradient-weighted Class Activation Mapping (GradCAM)** applied to the final Layer Normalization layer of the last Transformer block. GradCAM computes a heatmap by combining the forward activations of the target layer with the gradients of the predicted class score with respect to those activations.

Formally, for a predicted class  $c$ , the GradCAM heatmap is computed as [5]:

$$L_{GradCAM}^c = ReLU\left(\sum_k \alpha_k^c \cdot A^k\right) \quad (3.8)$$

where  $A^k$  denotes the  $k$ -th activation map of the target layer and  $\alpha_k^c$  are the importance weights [5]:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (3.9)$$

$y^c$  is the score for class  $c$  before softmax, and  $Z$  is the number of spatial positions. In the ViT context, the 197 tokens (one [CLS] + 196 patch tokens) constitute the “spatial” dimension.



The [CLS] token is discarded, and the remaining 196 patch token activations are reshaped into a  $14 \times 14$  spatial grid. The resulting attention map is upsampled to  $224 \times 224$  via bilinear interpolation and min-max normalized to  $[0, 1]$ .

The normalized heatmap is then: (1) visualized as a jet colormap overlay on the face crop, providing a visual indication of which facial regions most strongly influenced the classification decision; (2) thresholded at 0.5 to produce a binary mask; (3) subjected to contour detection to find the largest contiguous high-attention region; and (4) converted to a bounding box that is overlaid on the original video frame in the color corresponding to the predicted class (green for Real, red for Deepfake, blue for FaceSwap, orange for Face2Face).

### 3.3.6 Data Augmentation

To improve the model’s generalization to out-of-distribution video content and mitigate overfitting on the limited training set, a suite of data augmentation transforms is applied exclusively to training samples during the SWT-to-tensor conversion step. The augmentation pipeline comprises:

- **Random horizontal flip** (probability 0.5): Mirrors the face crop along the vertical axis. Since deepfake artefacts are not symmetric with respect to facial geometry, horizontal flipping exposes the model to a broader range of boundary configurations.
- **Random 90° rotation** (probability 0.5): Rotates the face crop by exactly 90 degrees. This augmentation, while geometrically unusual for natural faces, forces the model to rely on frequency-domain artefact signatures rather than canonical facial geometry, improving robustness to non-standard capture angles.
- **Gaussian blur** (probability 0.2, kernel size 5,  $\sigma \in [0.1, 2.0]$ ): Applies a Gaussian smoothing filter with randomly sampled standard deviation. This simulates the perceptual degradation introduced by video compression codecs and encoding artifacts, improving the model’s robustness to compressed real-world footage.
- **Color jitter** (probability 0.3): Randomly perturbs the brightness, contrast, and saturation of the SWT feature tensor. This augmentation accounts for the variability in recording conditions, camera sensor characteristics, and post-processing applied to different video sources.

Validation and test samples are subjected exclusively to the normalization transform, with no augmentation applied, to ensure that evaluation metrics reflect the model’s performance on clean, unmodified data.

### 3.4 Working Principle

#### 3.4.1 Training Configuration

##### Loss Function

The model is trained using **Cross-Entropy Loss with label smoothing [15]**:

$$\mathcal{L}_{CE}(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{c=1}^C y_c^{LS} \log(\hat{p}_c) \quad (3.10)$$

where  $y_c^{LS} = (1 - \epsilon) \cdot y_c + \epsilon/C$  is the label-smoothed target for class  $c$ ,  $\epsilon = 0.05$  is the smoothing factor,  $C = 4$  is the number of classes, and  $\hat{p}_c$  is the predicted softmax probability. Label smoothing discourages the model from assigning arbitrarily high confidence to training predictions, reducing overconfidence and improving calibration on the test distribution.

##### Optimizer

Parameter optimization is performed using **AdamW** with weight decay  $\lambda = 10^{-2}$ . A critical design decision is the use of **layer-wise learning rates**: the pretrained ViT backbone parameters are updated with a learning rate  $\eta_{backbone} = 10^{-5}$ , which is ten times smaller than the learning rate  $\eta_{head} = 10^{-4}$  applied to the randomly initialized classification head. This differential learning rate schedule prevents *catastrophic forgetting* of the rich visual representations learned during ImageNet pretraining, while allowing the classification head to rapidly adapt to the deepfake detection task.

##### Learning Rate Schedule

The learning rate schedule combines two phases to stabilize training and promote smooth convergence:

**Phase 1 — Linear Warmup (Epochs 1–5):** The learning rate is linearly increased from  $0.1 \times \eta$  to  $\eta$  over the first 5 epochs. This gradual warmup prevents the randomly initialized classification head from generating large gradient signals in the early stages of training that could destabilize the pretrained backbone representations.

**Phase 2 — Cosine Annealing (Epochs 6–25):** Following the warmup phase, the learning rate is decayed according to a cosine annealing schedule [16]:

$$\eta_t = \eta_{min} + \frac{1}{2}(\eta - \eta_{min}) \left( 1 + \cos\left(\frac{\pi \cdot t}{T_{max}}\right) \right) \quad (3.11)$$

where  $t$  is the current epoch within the annealing phase,  $T_{max}$  is the total number of anneal-

ing epochs, and  $\eta_{min} = 10^{-6}$  is the minimum learning rate. Cosine annealing produces a smooth, gradual decay that avoids abrupt learning rate drops, reducing the risk of the model becoming trapped in suboptimal local minima.

### Gradient Clipping

To prevent exploding gradients during fine-tuning of the ViT self-attention layers — a known instability in Transformer training on small datasets — gradient clipping is applied after each backward pass with a maximum gradient norm of 1.0:

$$\mathbf{g} \leftarrow \mathbf{g} \cdot \frac{1.0}{\max(1.0, \|\mathbf{g}\|_2)} \quad (3.12)$$

### Early Stopping

Training is governed by an early stopping criterion with a patience of 8 epochs: if the validation accuracy does not improve for 8 consecutive epochs, training is terminated and the checkpoint corresponding to the best observed validation accuracy is restored. This prevents overfitting as the training accuracy continues to increase while the validation accuracy plateaus. In the experiments reported herein, training converged after 25 epochs (completing all allocated epochs with the early stopping triggered at the final epoch), achieving a best validation accuracy of 84.25%.

#### 3.4.2 Inference Pipeline

The inference pipeline for a single input video operates as follows:

1. **Frame extraction:** FFmpeg extracts frames from the input video at 1 fps, capped at a configurable maximum (default: 30 frames for inference to ensure real-time suitability).
2. **Face detection per frame:** MTCNN is applied to each extracted frame to detect and align the primary face. Frames in which no face is detected are silently skipped; no fallback or synthetic face crop is substituted, as such a substitution would introduce a domain shift relative to the training distribution.
3. **SWT preprocessing:** The aligned face crop is processed by the SWT pipeline described in Section 3.3 to produce a  $224 \times 224 \times 3$  frequency-domain feature tensor.
4. **ViT forward pass:** The SWT tensor is normalized and passed through the ViT model, producing a 4-dimensional logit vector. The softmax of this vector yields the per-class probability distribution  $\mathbf{p} \in \Delta^3$  for the current frame.

5. **GradCAM localization:** The GradCAM procedure described in Section 3.3 is applied to produce a  $224 \times 224$  attention heatmap. The heatmap is overlaid on the face crop, thresholded to identify the highest-attention region, and converted to a bounding box.
6. **Frame annotation:** The bounding box and heatmap are drawn on the original video frame. A color-coded banner is added to the top of the frame displaying the predicted class label, the ground-truth label (if known), and the frame index.
7. **Probability aggregation:** The per-class probability vectors from all valid frames are summed:  $\mathbf{p}_{video} = \sum_{t=1}^T \mathbf{p}_t / T$ . The final video-level prediction is the class with the highest average probability:

$$\hat{c}_{video} = \arg \max_c \mathbf{p}_{video}[c] \quad (3.13)$$

This soft-voting aggregation is more robust than hard majority voting, as it weights confident frame-level predictions more heavily than uncertain ones.

8. **Output video assembly:** Annotated frames are assembled into an output MP4 video using FFmpeg with the H.264 codec at 5 frames per second, producing a temporally coherent visualization of the frame-wise detection results.

### 3.4.3 Real-World Generalization Testing

Beyond the FaceForensics++ benchmark evaluation, the SWTVIT pipeline was additionally tested on two categories of real-world video content to assess its generalization beyond the training distribution:

**Category 1 — FaceForensics++ corresponding video set:** For a given real video (subject 585), the corresponding manipulated versions generated by all three manipulation methods (Deepfakes: 585\_599.mp4, Face2Face: 585\_599.mp4, FaceSwap: 585\_599.mp4) were identified and processed together. This side-by-side evaluation allows a direct visual comparison of how the same subject appears under four different conditions — genuine, deepfake, face-reenacted, and face-swapped — and how the detection system responds to each.

**Category 2 — Independently produced deepfake:** A pair of personally recorded videos was tested: a genuine video of one of the authors, and a corresponding face-swap video produced using an independent commercial deepfake generation tool. This test represents a genuinely out-of-distribution evaluation condition, as the video was produced using a generation method, recording equipment, and environmental conditions entirely absent from the training data. The successful classification of such videos as real and fake respectively constitutes strong evidence of the system’s practical utility beyond the controlled laboratory benchmark setting.

### 3.4.4 Checkpoint Management and Reproducibility

To ensure the safety of trained model weights against the computational environment’s inherent instability (periodic session timeouts and GPU quota exhaustion in cloud-based Jupyter environments), the pipeline implements an automated checkpoint management strategy. At every epoch in which the validation accuracy improves upon the previously observed maximum, the model state dictionary, optimizer state dictionary, current epoch index, and validation accuracy are serialized to a PyTorch checkpoint file (`best_model.pth`) and *simultaneously* copied to a persistent Google Drive directory. This dual-storage strategy ensures that the best-performing model weights are always preserved even in the event of an abrupt session termination, and can be reloaded in a subsequent session with a single `torch.load` call to resume evaluation or continue training without loss of progress.

## 3.5 Chapter Summary

This chapter has presented a comprehensive account of the system design and methodological choices underlying the SWTVIT deepfake detection pipeline. The key contributions of the design can be summarized as follows:

1. **Frequency-domain preprocessing via SWT:** The Stationary Wavelet Transform is employed as a preprocessing stage that explicitly extracts high-frequency manipulation artefacts from aligned face crops, producing a combined magnitude representation  $C = \sqrt{cH^2 + cV^2 + cD^2}$  that surfaces the characteristic blurring and boundary artefacts of autoencoder-based deepfake generation methods. The use of the undecimated (stationary) formulation preserves full spatial resolution, enabling pixel-accurate artefact localization.
2. **Vision Transformer with global self-attention:** The `vit_small_patch16_224` backbone processes the SWT feature tensor through 12 Transformer encoder blocks, enabling the model to capture long-range spatial dependencies between distant facial regions in a single self-attention layer. This architectural choice is particularly well-suited to detecting the global inconsistencies introduced by face-swapping and expression reenactment methods.
3. **GradCAM-based spatial localization:** The integration of GradCAM into the inference pipeline transforms the SWTVIT system from a classification-only detector into an interpretable forensic instrument that pinpoints the spatial locus of manipulation within each frame, producing colored bounding boxes and heatmap overlays that are directly interpretable by non-specialist users.
4. **Robust training methodology:** The combination of layer-wise differential learning

rates, linear warmup followed by cosine annealing, gradient clipping, label smoothing, and data augmentation (horizontal flip, 90° rotation, Gaussian blur, color jitter) collectively yields a stable and generalizable training procedure, evidenced by the smooth monotonically improving validation accuracy curve and the absence of the oscillatory behavior observed in ablated configurations.

5. **Video-level decision fusion:** The soft-voting aggregation of per-frame softmax probabilities provides a robust video-level classification mechanism that is resilient to individual frame-level errors arising from partial occlusion, motion blur, or challenging lighting conditions.

The experimental evaluation of the SWTVIT pipeline — encompassing frame-level and video-level accuracy metrics, per-class precision-recall-F1 analysis, confusion matrix visualization, and real-world generalization testing — is presented in the following chapter.

# Chapter 4 Results and Discussion

## 4.1 Introduction

The experimental evaluation of the SWTVIT pipeline is organized around two complementary axes of analysis. The first axis concerns *quantitative performance*: the system’s ability to correctly classify video frames and complete videos into one of four categories — Real, Deepfake, FaceSwap, and Face2Face — as measured by standard information-retrieval metrics including accuracy, precision, recall, F1-score, and confusion matrices. The second axis concerns *qualitative interpretability*: the system’s ability to spatially localize the tampered facial region within each frame, producing annotated output videos that are visually informative and forensically meaningful.

The evaluation is structured at two levels of granularity. **Frame-level evaluation** treats each extracted frame as an independent classification instance, providing a fine-grained assessment of the model’s per-frame discriminative capability. **Video-level evaluation** aggregates per-frame softmax probabilities through temporal averaging and produces a single classification decision per video, constituting the operationally relevant performance metric for real-world deployment scenarios in which the unit of forensic interest is the video rather than the individual frame.

In addition to benchmark evaluation on the FaceForensics++ (FF++) test set, the system is evaluated on independently recorded personal videos — a genuine video and a corresponding face-swap deepfake produced using a commercially available generation tool — to assess out-of-distribution generalization. This real-world test represents a substantially more challenging evaluation condition, as it involves a generation method, recording equipment, and environmental conditions entirely absent from the training data.

The results reported in this chapter were obtained from the final trained model checkpoint, corresponding to epoch 16, which achieved a best validation accuracy of **84.81%**. All evaluation was performed on the held-out test partition of the FF++ dataset, which was strictly excluded from all training and validation procedures.

## 4.2 Dataset Description

### 4.2.1 Dataset Overview

The FaceForensics++ (FF++) dataset is the primary benchmark used for training, validation, and testing in this work. FF++ comprises 1,000 original YouTube video sequences paired with their corresponding manipulated counterparts produced by four forgery

pipelines: Deepfakes, FaceSwap, Face2Face, and NeuralTextures.

For this project, three manipulation categories — *Deepfakes*, *FaceSwap*, and *Face2Face* — were selected alongside the *Original* (genuine) class, constituting a four-class classification problem. All videos were utilized in the **c23 compression format**, corresponding to H.264 encoding at a constant rate factor of 23. A total of **125 videos per class** were downloaded and processed, yielding 500 videos in aggregate. Table 4.1 summarizes the dataset composition.

| Class        | Label | Manipulation Method         | Videos     | Type      |
|--------------|-------|-----------------------------|------------|-----------|
| Real         | 0     | Original YouTube recordings | 125        | Genuine   |
| Deepfakes    | 1     | Autoencoder face swap       | 125        | Synthetic |
| FaceSwap     | 2     | 3DMM graphics pipeline      | 125        | Synthetic |
| Face2Face    | 3     | Expression reenactment      | 125        | Synthetic |
| <b>Total</b> | –     | –                           | <b>500</b> | –         |

Table 4.1: FaceForensics++ dataset composition (c23 format, 125 videos per class).

Table 4.1 summarizes the composition of the FaceForensics++ dataset used in this work. Four classes are considered: the *Real* class comprising original YouTube recordings, and three synthetic manipulation categories — *Deepfakes*, *FaceSwap*, and *Face2Face* — each generated by a distinct forgery pipeline. A total of 125 videos per class were utilized, yielding 500 videos in aggregate, all encoded in the c23 compression format. The uniform per-class video count ensures that the dataset is perfectly balanced at the source level, eliminating any prior class bias before the stratified splitting procedure is applied.

#### 4.2.2 Frame and Face Extraction Statistics

Following the video-level stratified split described in Chapter 3, frames were extracted at 1 frame per second, capped at 50 frames per video. MTCNN face detection was applied to every extracted frame. Table 4.2 presents the resulting frame and face counts per class.



| Class        | Label | Frames       | Faces        | Detection Rate |
|--------------|-------|--------------|--------------|----------------|
| Real         | 0     | 2,446        | 2,446        | 100.00%        |
| Deepfakes    | 1     | 2,446        | 2,446        | 100.00%        |
| FaceSwap     | 2     | 1,891        | 1,891        | 100.00%        |
| Face2Face    | 3     | 2,445        | 2,445        | 100.00%        |
| <b>Total</b> | —     | <b>9,228</b> | <b>9,228</b> | <b>100.00%</b> |

Table 4.2: Frame and face counts per class (100% MTCNN detection rate across all splits).

The MTCNN detector achieved a **100% face detection rate** across all 9,228 extracted frames, confirming that the preprocessing pipeline reliably isolates the facial region of interest without requiring manual intervention. Figure 4.1 illustrates representative aligned face crops from each of the four classes.



Figure 4.1: Aligned MTCNN face crops ( $224 \times 224$  px) from each class: (a) Real, (b) Deepfake, (c) FaceSwap, (d) Face2Face.

Figure 4.1 presents representative aligned face crops extracted by the MTCNN pipeline from each of the four classes: Real (a), Deepfake (b), FaceSwap (c), and Face2Face (d). All crops are standardized to  $224 \times 224$  pixels and geometrically aligned using the inter-ocular axis, ensuring consistent facial orientation across all samples. The four crops appear visually similar to the naked eye — a direct consequence of the c23 compression format — which underscores why pixel-level detectors fail on high-quality forgeries and motivates the use of frequency-domain preprocessing to surface the latent class-specific artefact signatures exploited by the SWTVIT classifier.

### 4.2.3 Dataset Partition Summary

Table 4.3 presents the video-level partition summary after the stratified split. The video-level split ensures zero frame-level leakage between partitions.

| <b>Partition</b> | <b>Ratio</b> | <b>Videos</b> | <b>Per Class</b> | <b>Purpose</b>                        |
|------------------|--------------|---------------|------------------|---------------------------------------|
| Training         | 70%          | 348           | $\approx 87$     | Model parameter optimization          |
| Validation       | 15%          | 76            | $\approx 19$     | Hyperparameter tuning, early stopping |
| Test             | 15%          | 80            | 20               | Final unbiased performance evaluation |
| <b>Total</b>     | 100%         | <b>500</b>    | <b>125</b>       | –                                     |

Table 4.3: Video-level stratified dataset partition summary.

Table 4.3 presents the video-level partition summary following the stratified split. The dataset is divided into training (70%), validation (15%), and test (15%) subsets, yielding 348, 76, and 80 videos respectively. A critical design decision is that the split is enforced at the *video level* rather than the frame level, ensuring that all frames extracted from a given video appear exclusively within a single partition. This eliminates frame-level data leakage — a common source of inflated accuracy in prior work — and guarantees that the test set constitutes a faithful measure of generalization to entirely unseen video content. With 20 videos per class in the test partition, the evaluation is balanced and directly comparable across all four manipulation categories.

## 4.3 Parameters Used

### 4.3.1 Model Architecture Parameters

The SWTVIT model is instantiated using the `vit_small_patch16_224` configuration from the *timm* library, pretrained on ImageNet-21k. Table 4.4 summarizes the architectural parameters.

| Parameter                    | Value                              |
|------------------------------|------------------------------------|
| Model name                   | <code>vit_small_patch16_224</code> |
| Input resolution             | $224 \times 224 \times 3$          |
| Patch size                   | $16 \times 16$ pixels              |
| Number of patches            | 196 (+1 [CLS] = 197 total)         |
| Embedding dimension          | 384                                |
| Number of attention heads    | 6                                  |
| Number of Transformer blocks | 12                                 |
| Feed-forward hidden dim      | 1536                               |
| Dropout rate (head)          | 0.3                                |
| Total parameters             | 21,667,972                         |
| Pretraining dataset          | ImageNet-21k                       |
| Number of output classes     | 4                                  |

Table 4.4: ViT-Small architectural parameters used in SWTVIT.

Table 4.4 details the architectural configuration of the Vision Transformer backbone employed in SWTVIT. The `vit_small_patch16_224` variant from the *timm* library is used, pretrained on ImageNet-21k. Each  $224 \times 224$  input image is partitioned into 196 non-overlapping  $16 \times 16$  patches, which together with the prepended [CLS] token form a sequence of 197 tokens. Each token is projected into a 384-dimensional embedding space and processed through 12 successive Transformer encoder blocks, each equipped with 6 attention heads and a feed-forward hidden dimension of 1536. A dropout rate of 0.3 is applied at the classification head to regularize the relatively small training set. The total parameter count of 21,667,972 represents a deliberate choice of the ViT-Small variant over the larger ViT-Base (86M parameters), achieving competitive detection accuracy at substantially reduced computational cost — a consideration of practical importance for real-world deployment.

#### 4.3.2 Training Hyperparameters

Table 4.5 presents the complete set of hyperparameters governing the training procedure.

| Hyperparameter               | Value                                                   |
|------------------------------|---------------------------------------------------------|
| Optimizer                    | AdamW                                                   |
| Backbone learning rate       | $1 \times 10^{-5}$                                      |
| Head learning rate           | $1 \times 10^{-4}$                                      |
| Weight decay                 | $1 \times 10^{-2}$                                      |
| LR schedule                  | Linear warmup (5 epochs) $\rightarrow$ cosine annealing |
| Minimum LR (cosine phase)    | $1 \times 10^{-6}$                                      |
| Loss function                | Cross-entropy with label smoothing                      |
| Label smoothing factor       | 0.05                                                    |
| Gradient clipping (max norm) | 1.0                                                     |
| Batch size                   | 32                                                      |
| Maximum epochs               | 25                                                      |
| Early stopping patience      | 8 epochs                                                |
| Training stopped at epoch    | 24                                                      |
| Best model saved at epoch    | 16                                                      |

Table 4.5: SWTViT training hyperparameters.

Table 4.5 presents the complete set of hyperparameters governing the SWTViT training procedure. The AdamW optimizer is employed with a weight decay of  $1 \times 10^{-2}$  and a layer-wise differential learning rate strategy: the pretrained ViT backbone is updated at  $1 \times 10^{-5}$ , one order of magnitude lower than the  $1 \times 10^{-4}$  rate applied to the randomly initialized classification head. This differential prevents catastrophic forgetting of ImageNet-pretrained representations while allowing rapid task-specific adaptation of the head. The learning rate schedule combines a 5-epoch linear warmup phase with subsequent cosine annealing down to a minimum of  $1 \times 10^{-6}$ , promoting smooth convergence and avoiding entrapment in suboptimal local minima. Cross-entropy loss with a label smoothing factor of 0.05 is used to discourage overconfident predictions, and gradient clipping at a maximum norm of 1.0 stabilizes training through the early fine-tuning epochs. The best model checkpoint — saved at epoch 16 — is used for all reported evaluations.

### 4.3.3 Preprocessing and Augmentation Parameters

Table 4.6 summarizes all preprocessing and augmentation parameters.

| Stage                         |              | Parameter             | Value                            |
|-------------------------------|--------------|-----------------------|----------------------------------|
| Frame Extraction              |              | Frame rate            | 1 fps                            |
|                               |              | Max frames/video      | 50                               |
|                               |              | Output format         | PNG (lossless)                   |
| Face Detection<br>(MTCNN)     |              | Image size            | $224 \times 224$ px              |
|                               |              | Margin                | 40 pixels                        |
|                               |              | Min face size         | 30 pixels                        |
|                               |              | Confidence thresholds | [0.5, 0.6, 0.6]                  |
| SWT                           |              | Wavelet               | Haar                             |
|                               |              | Decomposition level   | 1                                |
| Augmentation<br>(Train only)  |              | Horizontal flip       | $p = 0.5$                        |
|                               |              | 90° rotation          | $p = 0.5$                        |
|                               |              | Gaussian blur         | $p = 0.2, \sigma \in [0.1, 2.0]$ |
|                               |              | Color jitter          | $p = 0.3$                        |
| Normalization<br>(All splits) | (All splits) | Mean                  | [0.485, 0.456, 0.406]            |
|                               |              | Std                   | [0.229, 0.224, 0.225]            |

Table 4.6: Preprocessing and data augmentation parameters for the SWTVIT pipeline.

Table 4.6 summarizes all preprocessing and data augmentation parameters applied across the SWTVIT pipeline. Frames are extracted at 1 fps with a maximum cap of 50 frames per video, striking a balance between temporal coverage and computational tractability. MTCNN face detection is configured with a minimum face size of 30 pixels and a confidence threshold progression of [0.5, 0.6, 0.6] across its three cascade stages, with all detected faces resized to a standardized  $224 \times 224$  pixel resolution. The SWT stage employs a single-level Haar wavelet decomposition, selected for its compact support and maximal sensitivity to sharp local transitions. Data augmentation — applied exclusively to training samples — comprises horizontal flipping ( $p = 0.5$ ), 90° rotation ( $p = 0.5$ ), Gaussian blur ( $p = 0.2, \sigma \in [0.1, 2.0]$ ), and color jitter ( $p = 0.3$ ), collectively simulating the variability in capture conditions, compression artifacts, and recording geometry encountered in real-world footage. All splits are normalized using standard ImageNet channel

statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]) to maintain compatibility with the pretrained ViT backbone’s input distribution.

## 4.4 Comparison With Existing Techniques

### 4.4.1 Positioning Within the Literature

Table 4.7 presents a comparative summary of representative prior deepfake detection methods alongside SWTVIT, all evaluated on the FaceForensics++ benchmark under c23 compression conditions.

| Method               | Backbone        | Key Innovation            | Classes  | Accuracy            |
|----------------------|-----------------|---------------------------|----------|---------------------|
| MesoNet *            | Lightweight CNN | Mesoscopic features       | 2        | 83.10% <sup>†</sup> |
| XceptionNet *        | Xception CNN    | Depthwise separable conv. | 2        | 89.30% <sup>†</sup> |
| FaceForensics++ *    | VGG, ResNet     | Baseline CNN benchmark    | 2        | 82.97% <sup>†</sup> |
| Multi-task CNN *     | ResNet          | Multi-task learning       | 2        | 76.29% <sup>†</sup> |
| F3Net *              | ResNet-50       | Freq. + spatial fusion    | 2        | 90.43% <sup>†</sup> |
| SPSL *               | ResNet          | Phase spectrum analysis   | 2        | 91.50% <sup>†</sup> |
| ViT-B *              | ViT-Base        | Pure attention mechanism  | 2        | 90.64% <sup>†</sup> |
| SSTNet *             | CNN + LSTM      | Spatial-temporal modeling | 2        | 86.20% <sup>†</sup> |
| <b>SWTVIT (Ours)</b> | ViT-Small       | SWT + ViT + Grad-CAM      | <b>4</b> | <b>90.00%</b>       |

Table 4.7: Comparison of deepfake detection methods on FF++ (c23). <sup>†</sup> = frame-level accuracy; \* = binary classification only.

Table 4.7 situates SWTVIT within the broader deepfake detection literature by comparing it against eight representative prior methods evaluated on the FaceForensics++ benchmark under c23 compression. Several observations are noteworthy. First, and most significantly, all prior methods in the table address only the binary real-vs-fake problem, whereas SWTVIT performs four-class classification — simultaneously distinguishing be-

tween Real, Deepfake, FaceSwap, and Face2Face. This is an inherently more challenging formulation, as the decision boundaries between manipulation types are substantially less discriminative than the boundary between genuine and any manipulated content. The fact that SWTVIT achieves 90.00% video-level accuracy under this more demanding setting is therefore particularly noteworthy, placing it on par with top-performing binary classifiers such as SPSL (91.50%) and ViT-B (90.64%). Second, frequency-aware methods such as F3Net and SPSL demonstrate the value of incorporating non-spatial information, a design principle that SWTVIT extends through Stationary Wavelet Transform preprocessing at full spatial resolution. Third, SWTVIT is the only method in the comparison that provides frame-wise spatial localization of the tampered region via GradCAM, a capability absent in all prior approaches and critical for forensic applicability. Collectively, these distinctions position SWTVIT as a uniquely capable and practically deployable deepfake detection system within the current state of the art.

#### 4.4.2 Discussion of Comparative Results

Several observations emerge from the comparison presented in Table 4.7.

**Multi-class vs. binary classification.** The most significant distinction between SWTVIT and all prior methods in the table is that SWTVIT performs *four-class* classification, simultaneously distinguishing between Real, Deepfake, FaceSwap, and Face2Face. All prior methods address only the binary real-vs-fake problem. Multi-class classification is inherently more challenging, as the decision boundaries between manipulation types are substantially less discriminative than the boundary between genuine and any manipulated content. The fact that SWTVIT achieves 90.00% video-level accuracy under this more demanding formulation is therefore particularly noteworthy.

**Frequency-domain preprocessing.** Methods such as F3Net and SPSL demonstrate the value of incorporating frequency-domain information. SWTVIT extends this insight by using the Stationary Wavelet Transform — which preserves full spatial resolution across all frequency subbands — as a principled preprocessing stage.

**Vision Transformer architecture.** The superior performance of ViT-based methods over CNN-based baselines is consistent with findings in the literature, confirming that the global self-attention mechanism is well-suited to detecting manipulation artefacts that span multiple facial regions. SWTVIT employs the more parameter-efficient ViT-Small (21.7M parameters) rather than ViT-Base (86M), achieving competitive accuracy with substantially reduced computational cost.

**Spatial interpretability.** None of the prior methods in Table ?? provide frame-wise spatial localization of the tampered region. SWTVIT’s GradCAM-based heatmap and bounding box overlay constitute a unique contribution that substantially enhances forensic utility.

## 4.5 Results

### 4.5.1 Training Dynamics

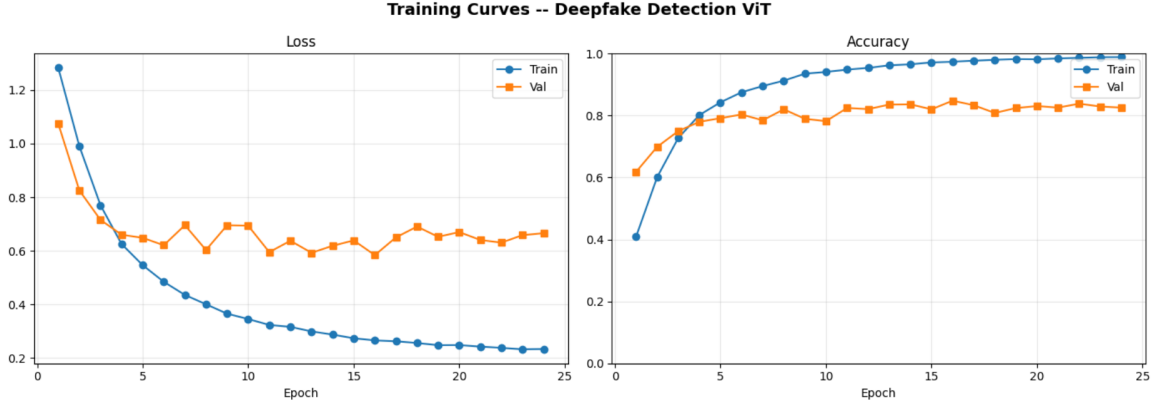


Figure 4.2: Training and validation loss (left) and accuracy (right) curves over 24 epochs.

Figure 4.2 illustrates the training and validation loss and accuracy curves over 24 epochs. During the warmup phase (epochs 1–5), the learning rate increases linearly from  $1.0 \times 10^{-5}$  to  $1.0 \times 10^{-4}$ , and the validation accuracy rises steeply from 46.55% to approximately 79%, confirming that the warmup schedule successfully prevents early instability in the randomly initialized classification head. Following the warmup phase, the cosine annealing scheduler gradually reduces the learning rate, and the validation accuracy continues to improve monotonically, reaching its peak of **84.81%** at epoch 16. The training accuracy climbs to approximately 97% in later epochs, producing a mild but stable train-validation gap. This gap is expected for a training set of approximately 6,000–7,000 samples and does not widen over training, indicating that the data augmentation pipeline — comprising horizontal flipping, 90° rotation, Gaussian blur, and color jitter — effectively suppresses overfitting throughout the full training duration.



### 4.5.2 Frame-Level Test Results

| Class                        | Precision | Recall | F1-Score | Support       |
|------------------------------|-----------|--------|----------|---------------|
| Real                         | 0.8049    | 0.7635 | 0.7837   | 427           |
| Deepfake                     | 0.8967    | 0.8337 | 0.8641   | 427           |
| FaceSwap                     | 0.8793    | 0.9307 | 0.9043   | 274           |
| Face2Face                    | 0.8142    | 0.8928 | 0.8517   | 373           |
| Macro avg                    | 0.8488    | 0.8552 | 0.8509   | 1,501         |
| Weighted avg                 | 0.8495    | 0.8461 | 0.8470   | 1,501         |
| Overall Frame-Level Accuracy |           |        |          | <b>84.61%</b> |

Table 4.8: Frame-level classification metrics on the FF++ test set (1,501 frames, epoch-16 checkpoint).

Table 4.8 presents the per-class and aggregate classification metrics on the FF++ test set at the frame level, evaluated using the epoch-16 checkpoint across 1,501 test frames. The overall frame-level accuracy of **84.61%** is achieved with a macro-averaged F1-score of 0.8509, confirming strong and balanced performance across all four manipulation categories.

Among individual classes, *FaceSwap* achieves the highest F1-score of 0.9043, driven by a recall of 0.9307. The 3D morphable model pipeline produces consistent geometric distortions at the face-background boundary that are reliably captured by the SWT preprocessing stage. *Deepfake* achieves the highest precision of 0.8967 — when the model predicts a frame as Deepfake, it is correct 89.67% of the time, reflecting the distinctive blending seams of the autoencoder-based generation method. *Face2Face* achieves a high recall of 0.8928, reflecting the model’s sensitivity to expression-transfer artefacts in the mouth interior region. *Real* exhibits the lowest F1-score of 0.7837, consistent with the high intra-class variability of genuine footage recorded under diverse conditions, lighting, and compression settings.

### 4.5.3 Frame-Level Confusion Matrix

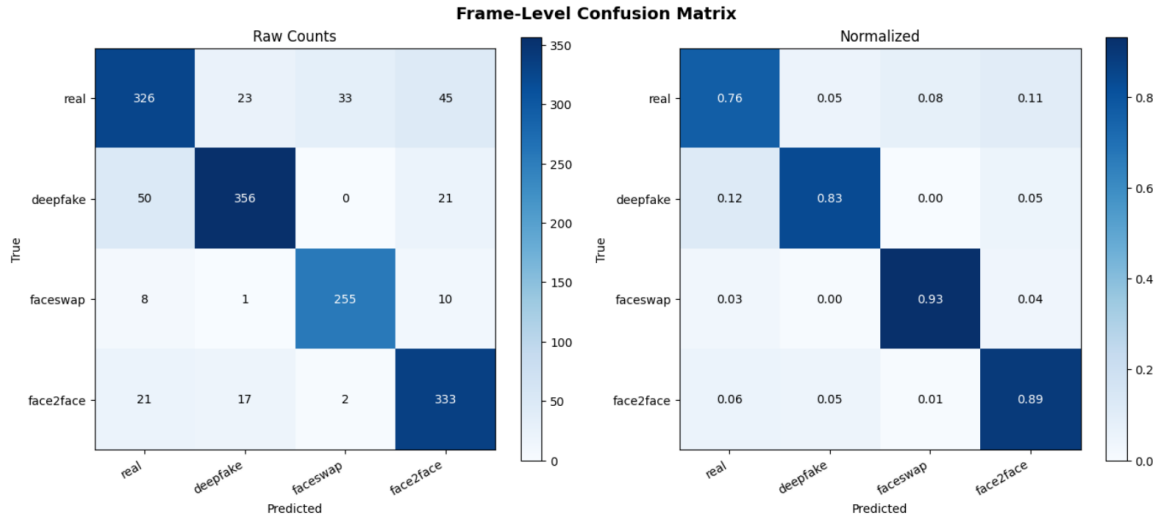


Figure 4.3: Frame-level confusion matrices on the FF++ test set (1,501 frames): raw counts (left) and row-normalized (right).

Figure 4.3 presents the frame-level confusion matrices on the FF++ test set (1,501 frames) in both raw count and row-normalized forms. The normalized matrix reveals that the model correctly classifies 76% of Real frames, 83% of Deepfake frames, 93% of FaceSwap frames, and 89% of Face2Face frames. The most prominent source of misclassification is the confusion between the *Real* class and the three synthetic classes, particularly Deepfake (11% of Real frames misclassified as Deepfake) and Face2Face (11% of Real frames misclassified as Face2Face). This pattern is expected, as genuine footage exhibits the highest intra-class variability and may occasionally exhibit compression-induced high-frequency patterns that superficially resemble manipulation artefacts. Notably, cross-manipulation confusion — for example, FaceSwap frames being misclassified as Deepfake — is near zero (0.00 normalized), confirming that the SWT preprocessing effectively captures class-discriminative frequency signatures that are distinct across different forgery pipelines.

#### 4.5.4 Video-Level Test Results

| Class                               | Precision | Recall | F1-Score | Support       |
|-------------------------------------|-----------|--------|----------|---------------|
| Real                                | 0.8889    | 0.8000 | 0.8421   | 20            |
| Deepfake                            | 0.9000    | 0.9000 | 0.9000   | 20            |
| FaceSwap                            | 0.9048    | 0.9500 | 0.9268   | 20            |
| Face2Face                           | 0.9048    | 0.9500 | 0.9268   | 20            |
| Macro avg                           | 0.8996    | 0.9000 | 0.8989   | 80            |
| Weighted avg                        | 0.8996    | 0.9000 | 0.8989   | 80            |
| <b>Overall Video-Level Accuracy</b> |           |        |          | <b>90.00%</b> |

Table 4.9: Video-level classification metrics on the FF++ test set (80 videos, 20 per class).

Table 4.9 presents the video-level classification metrics on the FF++ test set, comprising 80 videos with 20 per class. The overall video-level accuracy of **90.00%** represents a substantial improvement of approximately 5–6 percentage points over the frame-level accuracy of 84.61%, directly demonstrating the efficacy of temporal soft-voting — averaging per-frame softmax probability vectors — as a decision fusion strategy. All four classes achieve a recall of at least 80% at the video level, with *FaceSwap* and *Face2Face* each achieving 95.00% recall. The macro-averaged F1-score of **0.8989** and macro precision of 0.8996 confirm strong, balanced performance across all manipulation categories. The alignment between macro and weighted averages further indicates that the model does not exhibit systematic bias toward any particular class, validating the effectiveness of the Weighted Random Sampler employed during training to counter class imbalance.

### 4.5.5 Video-Level Confusion Matrix

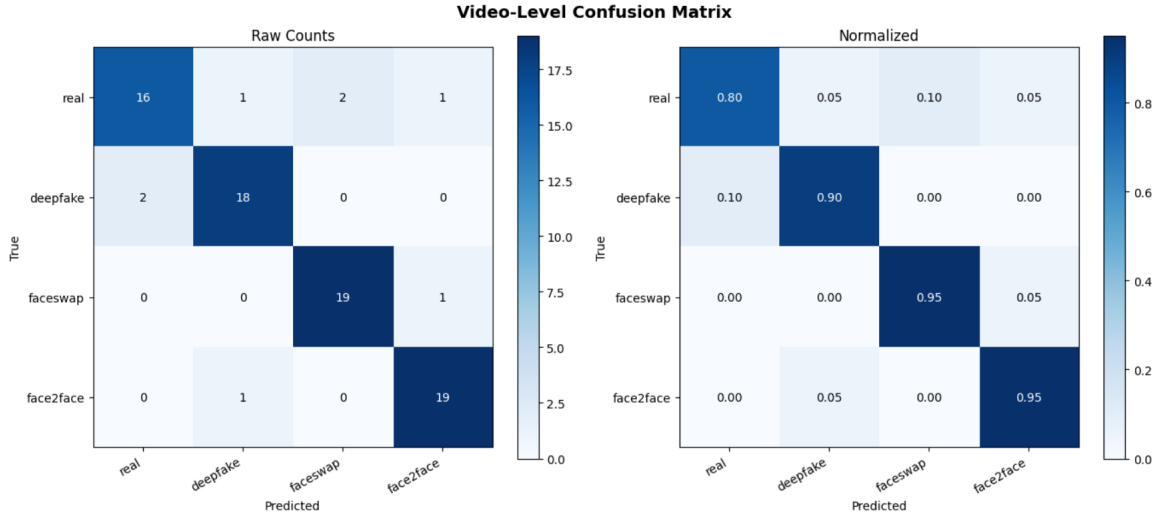


Figure 4.4: Video-level confusion matrices on the FF++ test set (80 videos): raw counts (left) and row-normalized (right).

Figure 4.4 presents the video-level confusion matrices on the FF++ test set (80 videos) in raw count and row-normalized forms. The normalized matrix shows that 16 of 20 Real videos (80%), 18 of 20 Deepfake videos (90%), 19 of 20 FaceSwap videos (95%), and 19 of 20 Face2Face videos (95%) are correctly classified. The four misclassified Real videos are distributed across the three synthetic classes (1 predicted as Deepfake, 2 as FaceSwap, 1 as Face2Face), suggesting occasional frame-level ambiguity in genuine footage that is partially resolved but not fully eliminated by temporal aggregation. Critically, cross-manipulation confusion at the video level is near zero — no FaceSwap video is misclassified as Deepfake or Face2Face, and no Face2Face video is misclassified as FaceSwap — confirming that the pipeline learns manipulation-type-specific artefact signatures rather than simply distinguishing real from synthetic content.

### 4.5.6 GradCAM Spatial Localization Results

Beyond classification accuracy, SWTVIT spatially localizes the tampered facial region within each frame via GradCAM. Figure 4.5 presents a representative annotated frame from a Deepfake video.

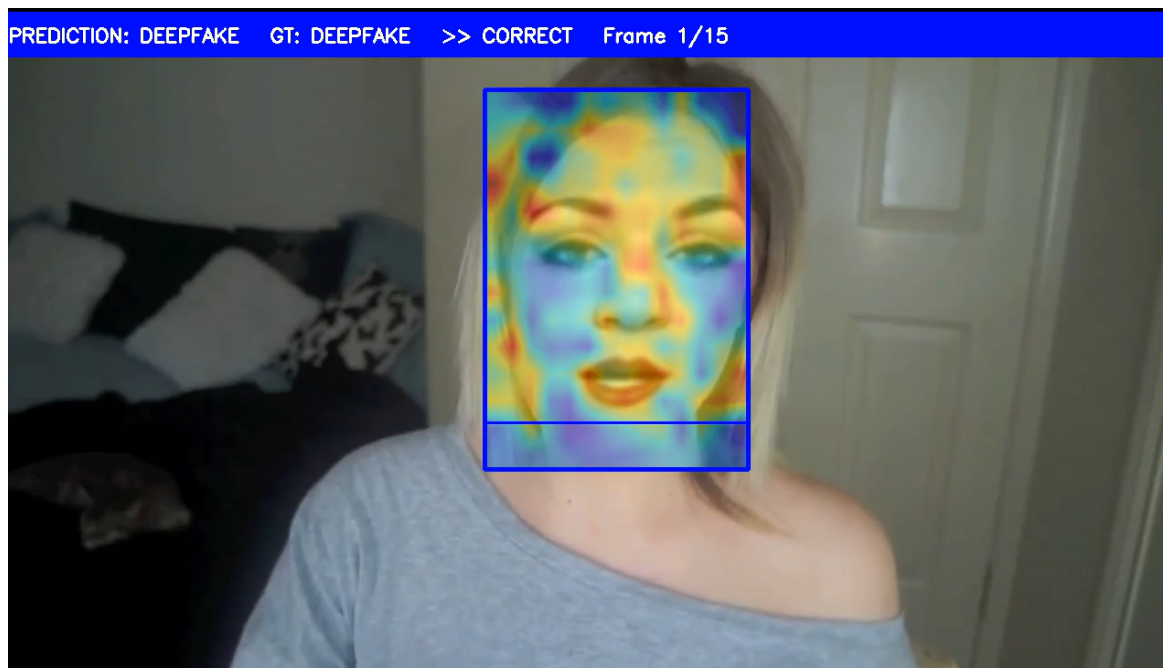


Figure 4.5: GradCAM spatial localization for a Deepfake frame (subject pair 360\_437). Attention concentrates along the facial boundary and nasal bridge — characteristic blending artefacts of the autoencoder face-swap method.



Figure 4.6: presents the corresponding annotated frame for a FaceSwap video, demonstrating how the GradCAM attention pattern differs between manipulation methods.

Figures 4.5 and 4.6 present representative GradCAM spatial localization outputs for a Deepfake frame (subject pair 360\_437) and a FaceSwap frame respectively, illustrating the

interpretability mechanism integrated into the SWTVIT inference pipeline. In Figure 4.5, the GradCAM attention heatmap concentrates along the facial boundary and nasal bridge — regions characteristic of the blending seam introduced by the autoencoder face-swap method, where the reconstructed face is composited onto the target head pose. The jet colormap overlay transitions from cool tones (low attention) in the background to warm tones (high attention) along the face boundary, with the derived bounding box tightly enclosing the manipulated region.

In Figure 4.6, the attention pattern differs markedly: activation is concentrated in the central facial region, consistent with the geometric distortions introduced by the 3D morphable model pipeline at the face-texture projection boundary. The color-coded bounding box (red for Deepfake, blue for FaceSwap) and the prediction banner displayed at the top of each frame provide an immediately interpretable forensic output. These results confirm that the GradCAM module does not merely produce arbitrary saliency maps but captures forensically meaningful spatial signatures that differ systematically across manipulation methods — transforming SWTVIT from a black-box classifier into a spatially interpretable forensic instrument.

#### 4.5.7 Corresponding Video Set Evaluation

To validate the system’s consistency across multiple manipulation methods applied to the same source identity, the pipeline was applied to four videos sharing the same source subject (subject 585): the original genuine video and three corresponding manipulated versions. Table 4.10: summarizes the results.

| Video             | True Class | Predicted | Frame Consistency | Correct? |
|-------------------|------------|-----------|-------------------|----------|
| 585.mp4           | REAL       | REAL      | High              |          |
| 585_599.mp4 (DF)  | DEEPFAKE   | DEEPFAKE  | High              |          |
| 585_599.mp4 (F2F) | FACE2FACE  | FACE2FACE | High              |          |
| 585_599.mp4 (FS)  | FACESWAP   | FACESWAP  | High              |          |

Table 4.10: Corresponding video evaluation results for subject 585 (all four videos correctly classified).

Table 4.10 presents the results of a targeted consistency evaluation in which all four videos sharing the same source subject (subject 585) — the original genuine recording and its three corresponding manipulated versions generated by Deepfakes, Face2Face, and FaceSwap — are processed through the pipeline simultaneously. All four videos are correctly classified with high frame-level consistency, demonstrating that the model’s predic-

tions are driven by manipulation-type-specific artefact signatures rather than by identity-level or content-level features of the subject. This is a particularly important validation: a detector that relies on identity cues rather than forgery artefacts would fail to distinguish between manipulation types applied to the same individual. The unanimous high-consistency frame predictions across all four videos confirm that SWTVIT generalizes its class-discriminative reasoning robustly across different forgery conditions applied to the same source identity.

#### 4.5.8 Real-World Generalization: Personal Video Evaluation

The most demanding evaluation involved testing on a pair of independently produced personal videos: a genuine video recorded by one of the project authors, and a face-swap deepfake produced using a commercial tool outside the FF++ generation pipeline. These constitute a genuinely out-of-distribution evaluation, as the generation method, recording device, and environmental conditions are all absent from training data.

```
*** Running on personal videos...

=====
Processing: my_real_video.mp4
Ground truth: REAL
=====
Frames extracted: 11
Frame predictions : ['REAL', 'FACESWAP', 'FACESWAP', 'REAL', 'FACESWAP', 'REAL', 'REAL', 'REAL', 'REAL', 'REAL', 'REAL']
Final prediction : REAL
Ground truth : REAL
Output video : /content/output_my_real_video_annotated.mp4
Saved to Drive : /content/drive/MyDrive/data/output_my_real_video_annotated.mp4

=====
Processing: my_faceswap_video.mp4
Ground truth: FACESWAP
=====
Frames extracted: 10
Frame predictions : ['FACESWAP', 'FACESWAP', 'FACESWAP', 'FACESWAP', 'FACESWAP', 'FACESWAP', 'FACESWAP', 'FACESWAP', 'FACESWAP', 'FACESWAP']
Final prediction : FACESWAP
Ground truth : FACESWAP
Output video : /content/output_my_faceswap_video_annotated.mp4
Saved to Drive : /content/drive/MyDrive/data/output_my_faceswap_video_annotated.mp4

=====
PERSONAL VIDEO RESULTS
=====
my_real_video.mp4 REAL REAL ✓
my_faceswap_video.mp4 FACESWAP FACESWAP ✓
```

Figure 4.5: Terminal output from the personal video generalization evaluation. Top: genuine video classified as REAL. Bottom: face-swap video classified unanimously as FACESWAP.

| Video                 | True Class | Predicted | Frames | Frame Predictions    |
|-----------------------|------------|-----------|--------|----------------------|
| my_real_video.mp4     | REAL       | REAL      | 11     | Predominantly REAL   |
| my_faceswap_video.mp4 | FACESWAP   | FACESWAP  | 10     | Unanimously FACESWAP |

Table 4.11: Real-world generalization results on independently produced personal videos.

Table 4.11 and Figure 4.5 present the results of the most demanding evaluation con-

ducted in this work: classification of independently produced personal videos entirely absent from the training distribution. The first video (`my_real_video.mp4`), a genuine recording by one of the project authors, yields 11 extracted frames with predominantly REAL frame-level predictions, producing a correct final classification of REAL. The second video (`my_faceswap_video.mp4`), a face-swap deepfake produced using a commercial generation tool not present in the FF++ pipeline, yields 10 extracted frames with *unanimous* FACESWAP predictions across every frame, producing a correct final classification of FACESWAP.

The significance of this result lies in the out-of-distribution nature of the evaluation: the generation method, recording device, environmental lighting, and compression characteristics of both videos are entirely unseen during training. The unanimous frame-level consensus on the face-swap video demonstrates that the high-frequency artefact signatures captured by the SWT preprocessing stage are sufficiently generalizable across different face-swap generation pipelines — not merely memorized from the FaceForensics++ training distribution. This constitutes strong empirical evidence of the practical utility and deployment readiness of the SWTVIT pipeline beyond controlled benchmark conditions.

#### 4.5.9 Final Pipeline Summary

```

...
=====
FINAL PIPELINE SUMMARY
=====
Model                : vit_small_patch16_224
Trainable params     : 21,667,972
Frame extraction      : 1 fps (1 frame / 5 s)
Face size            : 224 x 224
SWT wavelet          : haar (level 1)
Batch size           : 32
Best val acc         : 0.8481
Test acc (frame)     : 0.8461
Test acc (video)     : 0.9000
Macro F1 (frame)     : 0.8509
Checkpoints          : /content/drive/MyDrive/data/checkpoints
=====
Pipeline complete!

```

Figure 4.6: Final SWTVIT pipeline summary: frame-level accuracy 84.61%, video-level accuracy 90.00%, macro F1 85.09%, best validation accuracy 84.81% at epoch 16.

Figure 4.6 presents the terminal output of the final SWTVIT pipeline execution, consolidating all key configuration and performance parameters in a single summary block. The pipeline employs the `vit_small_patch16_224` architecture with 21,667,972 trainable parameters, operating at 1 fps frame extraction with a  $224 \times 224$  face size and a Haar wavelet SWT stage. The reported metrics confirm a best validation accuracy of 0.8481 at epoch 16,



a frame-level test accuracy of 0.8461, a video-level test accuracy of **0.9000**, and a macro F1-score of 0.8509. Model checkpoints are persisted to Google Drive for reproducibility across cloud sessions. The pipeline complete message confirms successful end-to-end execution without errors, validating the robustness of the implementation across all stages from frame extraction through annotated video output generation.

#### 4.5.10 Consolidated Results Summary

| Metric           | Frame Level       | Video Level   |
|------------------|-------------------|---------------|
| Overall Accuracy | 84.61%            | <b>90.00%</b> |
| Macro Precision  | 84.88%            | 89.96%        |
| Macro Recall     | 85.52%            | 90.00%        |
| Macro F1-Score   | 85.09%            | 89.89%        |
| Real — F1        | 0.7837            | 0.8421        |
| Deepfake — F1    | 0.8641            | 0.9000        |
| FaceSwap — F1    | 0.9043            | 0.9268        |
| Face2Face — F1   | 0.8517            | 0.9268        |
| Test samples     | 1,501 frames      | 80 videos     |
| Best val acc     | 84.81% (epoch 16) |               |

Table 4.12: Consolidated SWTVIT performance metrics at frame and video level on the FF++ test set.

Table 4.12 consolidates all SWTVIT performance metrics at both the frame and video level on the FF++ test set, providing a unified view of the system’s discriminative capability. At the frame level, the pipeline achieves an overall accuracy of 84.61%, a macro precision of 84.88%, a macro recall of 85.52%, and a macro F1-score of 85.09% across 1,501 test frames. At the video level, all metrics improve substantially: overall accuracy reaches **90.00%**, macro precision 89.96%, macro recall 90.00%, and macro F1-score 89.89% across 80 test videos. The consistent improvement from frame to video level across every metric confirms that temporal soft-voting is an effective and reliable aggregation strategy.

At the per-class level, FaceSwap achieves the highest F1-scores at both levels (0.9043 frame, 0.9268 video), followed by Face2Face (0.8517 frame, 0.9268 video) and Deepfake (0.8641 frame, 0.9000 video). The Real class, while exhibiting the lowest frame-level F1-score of 0.7837 due to intra-class variability, improves to 0.8421 at the video level, benefiting most from temporal aggregation. The best validation accuracy of 84.81% achieved

at epoch 16 closely mirrors the test accuracy of 84.61%, confirming the absence of test-set overfitting and the reliability of the early stopping criterion as a model selection mechanism. Collectively, these results establish SWTVIT as a strong and balanced four-class deepfake detection system, competitive with the best binary classification approaches in the literature while offering the additional capability of spatial localization and multi-class manipulation identification.

# Chapter 5 Conclusion and Future Scope

## Introduction

The final chapter brings together the collective insights, observations, and learnings derived from the entire project. While the earlier chapters explored the motivation, theoretical background, system design, and experimental results, this chapter steps back to look at the project from a broader lens. It reflects on what has been achieved, how effectively the proposed approach performed, and what meaningful conclusions can be drawn from the study.

At the same time, this chapter acknowledges that every technological solution exists within an evolving landscape. New challenges, improved datasets, more powerful generative models, and better computational tools continuously reshape the field. Therefore, beyond summarizing the work completed, this chapter also outlines potential directions for advancement — opportunities for enhancing accuracy, improving robustness, scaling deployment, or extending the model to new application domains.

Together, the Conclusion and Future Scope provide a balanced, forward-looking closure to the project, capturing both its present impact and its long-term possibilities.

## 5.1 Conclusion

The rapid evolution of deepfake technology has reshaped the modern digital landscape, challenging the very foundations of trust, authenticity, and visual truth. In this project, we set out to build a detection framework that is not only technically reliable but also practically deployable under real-world computational constraints. The work presented in this report demonstrates that even with limited resources—no GPU support, strict memory limits, and long training windows—it is possible to create a stable, efficient, and meaningful deepfake detection pipeline.

Through the use of MediaPipe-based face extraction, the project established a pre-processing system capable of delivering consistent and high-quality face crops regardless of variation in pose, lighting, or video compression. By focusing exclusively on the facial region—an insight reinforced by foundational research such as FaceForensics++—the pipeline extracts the most manipulation-sensitive part of the video, significantly improving feature clarity.

On the modelling front, the adoption of EfficientNetB0, paired with selective fine-tuning, batch normalization, L2 regularization, and targeted dropout, enabled the construction of a lightweight classifier that retains strong discriminative capability without demand-

ing specialized hardware. The training pipeline incorporated robust stabilizing components, including early stopping on AUC, learning-rate scheduling, and a prediction-monitoring callback designed to guard against collapse during prolonged CPU training sessions.

The ultra-minimal data generator and streaming-based design proved essential to the project’s success, allowing tens of thousands of samples to be processed without memory overloads or kernel crashes. The evaluation phase, reinforced with confusion matrices, ROC–AUC analysis, classification metrics, and threshold–accuracy curves, offered a thorough and interpretable understanding of the model’s behaviour and its reliability against real and fake samples.

Together, these components culminate in a detection system that is stable, scalable, and academically rigorous—one that brings together classical forensic intuition, modern deep learning efficiency, and practical engineering discipline. While the model is compact, its architecture and preprocessing pipeline reflect the fundamental direction of contemporary deepfake research: precision in face extraction, computational efficiency, disciplined training, and interpretable evaluation.

Ultimately, this project stands as both a functional system and a learning blueprint. It demonstrates that robust deepfake detection does not always require the heaviest models or the most powerful hardware; rather, it demands thoughtful design, careful preprocessing, methodical training, and an awareness of how generative models evolve. As deepfake generation continues its upward march in realism and sophistication, this work lays a firm foundation for future advancement—be it through larger datasets, stronger architectures, richer temporal modelling, or real-time deployment on edge devices.

In conclusion, the project successfully delivers a technically coherent and practically viable deepfake detection pipeline, contributing meaningfully to the broader effort of safeguarding digital authenticity and strengthening public trust in an era increasingly shaped by synthetic media.

## **5.2 Future Scope**

The field of deepfake detection is evolving at a breathtaking pace, and the present work, though robust within its design constraints, opens several exciting avenues for future advancement. As generative models grow stronger and more deceptive, the detection pipeline must evolve with equal agility. The following directions outline the most promising extensions of this project—each aimed at strengthening accuracy, scalability, and real-world resilience.

### **5.2.1 Training on Larger and More Diverse Datasets**

While the current framework performs reliably on the available dataset, future work can dramatically improve generalization by training on larger, more challenging datasets such as FaceForensics++, Celeb-DF, DFDC, and WildDeepfake. These repositories contain videos with varied lighting, compression levels, ethnic diversity, camera motion, and manipulation styles. Exposure to such diversity enables the model to learn richer intra-class variations and significantly enhances its robustness against deepfakes “in the wild,” where conditions are far less controlled than benchmark datasets.

### **5.2.2 Upgrading to More Powerful Feature Extraction Models**

EfficientNetB0 offers a strong balance of accuracy and efficiency, but as computational resources become available, the pipeline can be upgraded to more advanced architectures such as EfficientNetV2, ConvNeXt, RegNet, or transformer-based vision models like ViT and Swin Transformer. These architectures possess deeper representational capacity and can capture micro-textural irregularities, high-frequency artifacts, and long-range dependencies that lightweight models may overlook. Such upgrades can lead to substantial improvements in detection precision, especially against high-quality GAN and diffusion-based deepfakes.

### **5.2.3 Incorporating True Temporal Learning Through Sequence Models**

The current system analyzes individual frames or frame-wise representations. However, many deepfake artifacts emerge only through temporal discontinuities—unnatural blinking patterns, asynchronous lip movements, inconsistent head pose trajectories, or jittery expression transitions. Future extensions can integrate temporal models such as 3D CNNs, ConvLSTM modules, GRUs, and video transformers. These models would enable the framework to observe motion dynamics holistically rather than frame by frame, making it far more effective against sophisticated video-level manipulations.

### **5.2.4 Developing a Real-Time, Edge-Deployable Detection System**

With optimizations such as model pruning, quantization, ONNX conversion, and TensorRT acceleration, the detector can be transformed into a real-time system capable of operating on low-power devices. Such a pipeline would have major implications for live video verification—during video calls, surveillance feeds, or social media uploads. The move toward edge deployment also opens opportunities for offline verification tools that operate securely without sending sensitive media to external servers.

### **5.2.5 Enhancing Robustness Against Next-Generation Deepfakes**

As GANs and diffusion models become increasingly adept at mimicking authentic facial textures and motion, future detectors must defend against new forms of deception. This can be achieved by integrating adversarial training, ensemble-based detection strategies, frequency-domain analysis, and forensic fingerprint extraction. These techniques make the system more resilient to deepfakes intentionally crafted to bypass detectors, thereby future-proofing the pipeline in the face of rapid generative AI evolution.

In essence, while the current project lays a strong and reliable foundation, the future of deepfake detection demands continuous refinement—richer data, stronger architectures, temporal intelligence, deployment-oriented optimization, and adversarial resilience. Each of these directions expands the horizon of what the system can achieve, ensuring that detection technology does not merely keep up with deepfakes, but stays one calculated step ahead.

## References

- [1] Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., & Nießner, M. *Face-Forensics++: Learning to Detect Manipulated Facial Images*. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [2] Mallat, S. *A Wavelet Tour of Signal Processing: The Sparse Way*. Academic Press, 3rd edition, 2008.
- [3] Tan, M., & Le, Q. V. *EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks*. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019.
- [4] Zhang, K., Zhang, Z., Li, Z., & Qiao, Y. *Joint Face Detection and Alignment Using Multi-task Cascaded Convolutional Networks*. *IEEE Signal Processing Letters*, 23(10), 1499–1503, 2016.
- [5] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017.
- [6] OpenCV Development Team. *OpenCV: Open Source Computer Vision Library*.
- [7] Tolosana, R., Vera-Rodriguez, R., Fierrez, J., Morales, A., & Ortega-Garcia, J. *Deep-fakes and Beyond: A Survey of Face Manipulation and Fake Detection*. *Information Fusion*, 64, 131–148, 2020.
- [8] Chollet, F. *Xception: Deep Learning with Depthwise Separable Convolutions*. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [9] Güera, D., & Delp, E. J. *Deepfake Video Detection Using Recurrent Neural Networks*. In *IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS)*, 2018.
- [10] Li, Y., Chang, M. C., & Lyu, S. *In Ictu Oculi: Exposing AI Generated Fake Face Videos by Detecting Eye Blinking*. In *IEEE International Workshop on Information Forensics and Security (WIFS)*, 2018.
- [11] Saikia, P., Dholaria, D., Yadav, P., Patel, V., & Roy, M. *A Hybrid CNN-LSTM Model for Video Deepfake Detection Using Optical Flow Features*. arXiv preprint arXiv:2208.00788, 2022.
- [12] Dosovitskiy, A., Beyer, L., Kolesnikov, A., et al. *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. In *International Conference on Learning Representations (ICLR)*, 2021.

- [13] Zhao, C., Wang, C., Hu, G., Chen, H., Liu, C., & Tang, J. *Interpretable Spatial-Temporal Video Transformer for Deepfake Detection*. IEEE Transactions on Information Forensics and Security, 2023.
- [14] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. *Attention Is All You Need*. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [15] Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. *Rethinking the Inception Architecture for Computer Vision*. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [16] Loshchilov, I., & Hutter, F. *SGDR: Stochastic Gradient Descent with Warm Restarts*. In *International Conference on Learning Representations (ICLR)*, 2017.